

18. DMA Controller (DMACAa)

This MCU incorporates an 8-channel direct memory access controller (DMAC).

The DMAC is a module to transfer data without the CPU. When a DMA transfer request is generated, the DMAC transfers data stored at the transfer source address to the transfer destination address.

18.1 Overview

Table 18.1 lists the specifications of the DMAC, and Figure 18.1 shows a block diagram of the DMAC.

Table 18.1 Specifications of DMAC

Item		Description
Number of channels		8 (DMACm (m = 0 to 7))
Transfer space		512 Mbytes (0000 0000h to 0FFF FFFFh and F000 0000h to FFFF FFFFh excluding reserved areas)
Maximum transfer volume		64M data (Maximum number of transfers in block transfer mode: 1024 data × 65536 blocks)
DMA request source		- Request source selectable for each channel - Software trigger - Interrupt requests from peripheral modules or trigger input to external interrupt input pins*1
Channel priority		Channel 0 > Channel 1 > Channel 2 > Channel 3... > Channel 7 (Channel 0: Highest)
Transfer data	Single data	Bit length: 8, 16, 32 bits
	Block size	Number of data: 1 to 1024
Transfer mode	Normal transfer mode	- One data transfer by one DMA transfer request - Free running mode (setting in which total number of data transfers is not specified) settable
	Repeat transfer mode	- One data transfer by one DMA transfer request - Program returns to the transfer start address on completion of the repeat size of data transfer specified for the transfer source or destination. - Maximum settable repeat size: 1024
	Block transfer mode	- One block data transfer by one DMA transfer request - Maximum settable block size: 1024 data
Selective functions	Extended repeat area function	- Function in which data can be transferred by repeating the address values in the specified range with the upper bit values in the transfer address register fixed - Area of 2 bytes to 128 Mbytes separately settable as extended repeat area for transfer source and destination
Interrupt request	Transfer end interrupt	Generated when the specified number of transfers is completed in normal transfer mode Generated when the specified repeat count of transfers is completed in repeat transfer mode Generated when the specified block count of transfers is completed in block transfer mode
	Transfer escape end interrupt	Generated when the repeat size of data transfer is completed or the extended repeat area overflows.
Event link function		An event link request is generated after each data transfer (for block transfer, after each block is transferred).
Power consumption reduction function		Module-stop state can be set.

Note 1. For details on DMA request sources, refer to Table 15.5, Interrupt Vector Table in section 15, Interrupt Controller (ICUD).

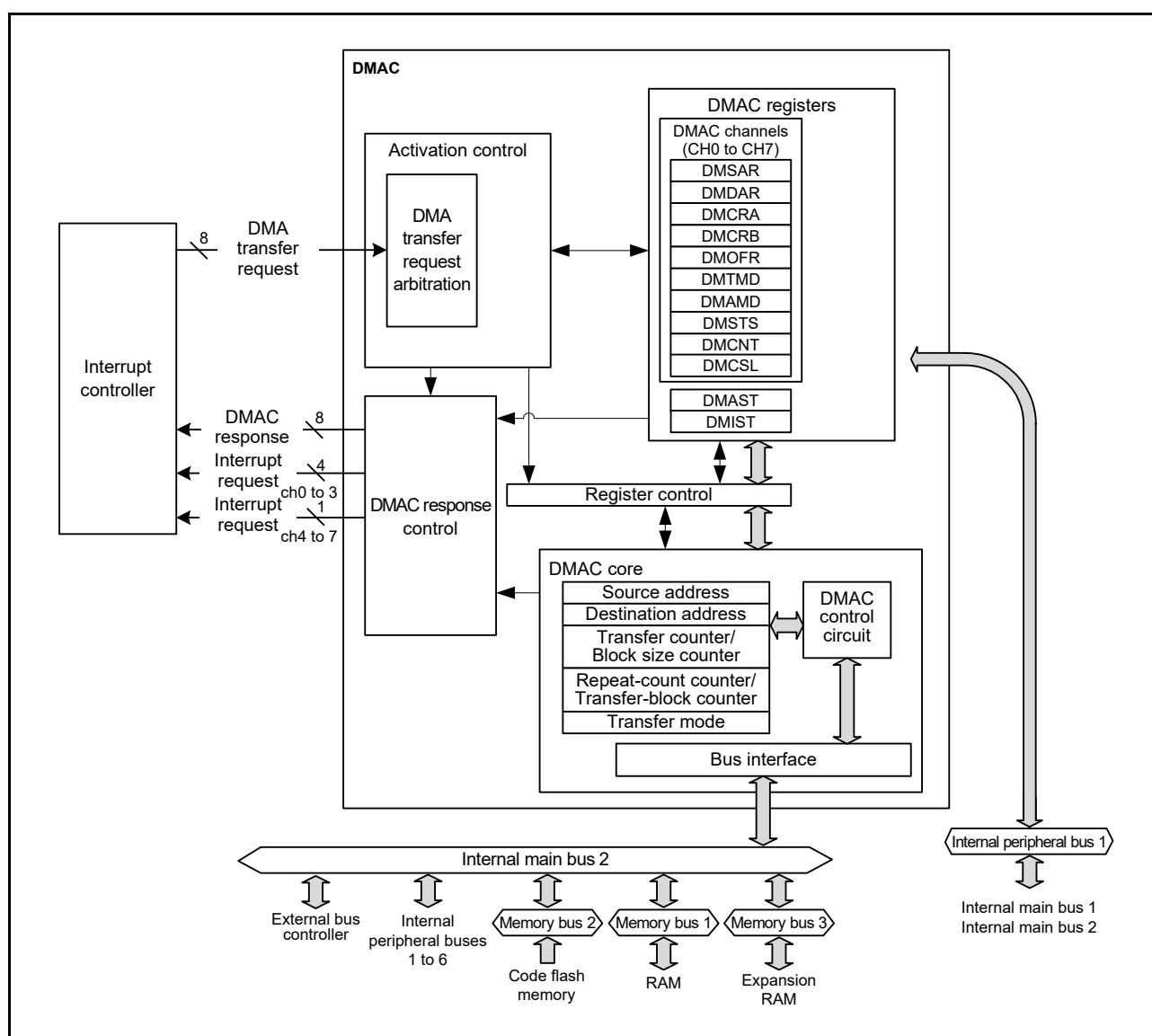
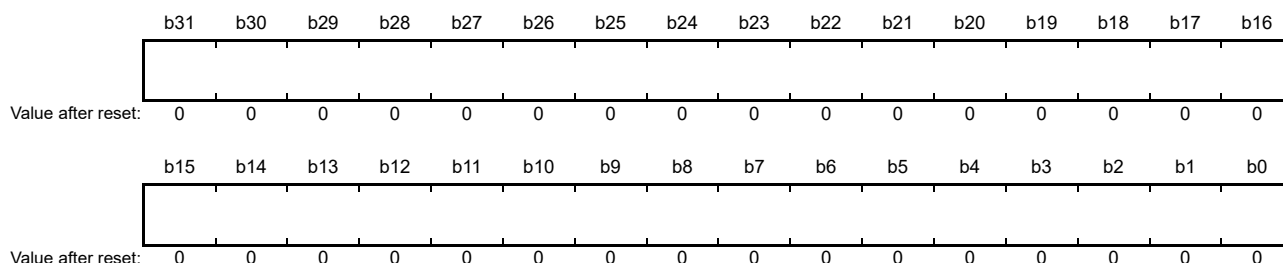


Figure 18.1 Block Diagram of DMAC

18.2 Register Descriptions

18.2.1 DMA Source Address Register (DMSAR)

Address(es): DMAC0.DMSAR 0008 2000h, DMAC1.DMSAR 0008 2040h, DMAC2.DMSAR 0008 2080h, DMAC3.DMSAR 0008 20C0h, DMAC4.DMSAR 0008 2100h, DMAC5.DMSAR 0008 2140h, DMAC6.DMSAR 0008 2180h, DMAC7.DMSAR 0008 21C0h



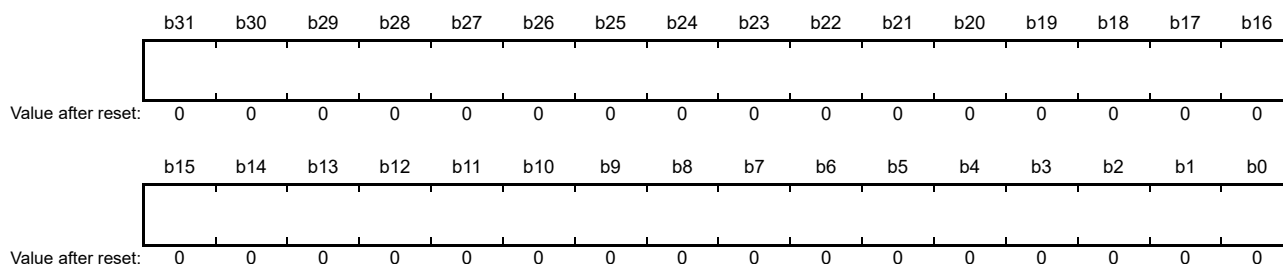
Bit	Description	Setting Range	R/W
b31 to b0	Specifies the transfer source start address.	0000 0000h to 0FFF FFFFh (256 Mbytes) F000 0000h to FFFF FFFFh (256 Mbytes)	R/W

Set the DMSAR register while DMAC stops (DMAST.DMST bit = 0) or DMA transfer is disabled (DMCNT.DTE bit = 0).

Setting bits 31 to 29 is invalid; a value of bit 28 is extended to bits 31 to 29. Reading the DMSAR register returns the extended value.

18.2.2 DMA Destination Address Register (DMDAR)

Address(es): DMAC0.DMDAR 0008 2004h, DMAC1.DMDAR 0008 2044h, DMAC2.DMDAR 0008 2084h, DMAC3.DMDAR 0008 20C4h, DMAC4.DMDAR 0008 2104h, DMAC5.DMDAR 0008 2144h, DMAC6.DMDAR 0008 2184h, DMAC7.DMDAR 0008 21C4h



Bit	Description	Setting Range	R/W
b31 to b0	Specifies the transfer destination start address.	0000 0000h to 0FFF FFFFh (256 Mbytes) F000 0000h to FFFF FFFFh (256 Mbytes)	R/W

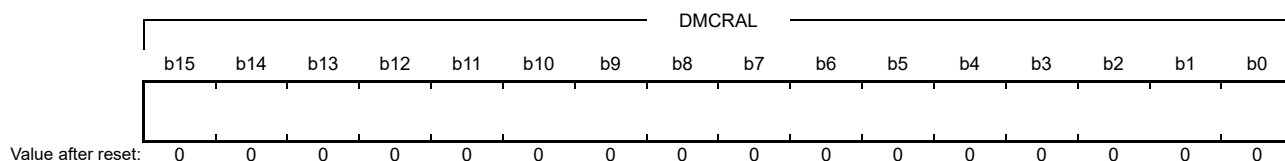
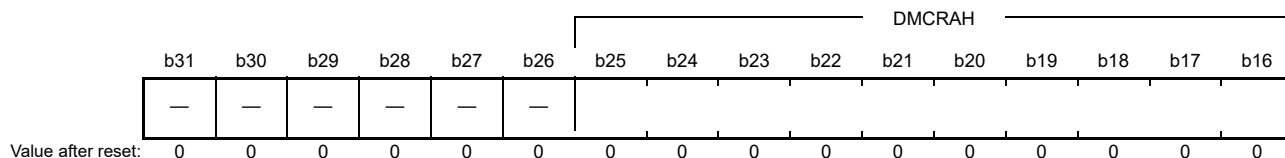
Set the DMDAR register while DMAC stops (DMAST.DMST bit = 0) or DMA transfer is disabled (DMCNT.DTE bit = 0).

Setting bits 31 to 29 is invalid; a value of bit 28 is extended to bits 31 to 29. Reading the DMDAR register returns the extended value.

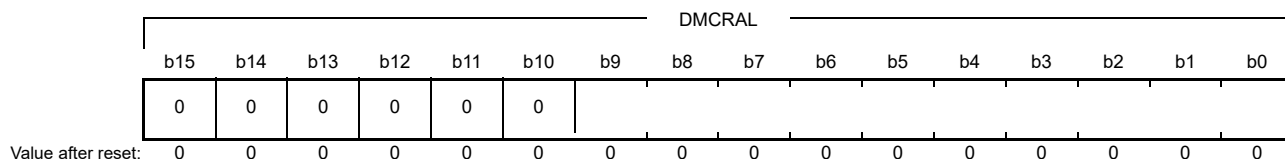
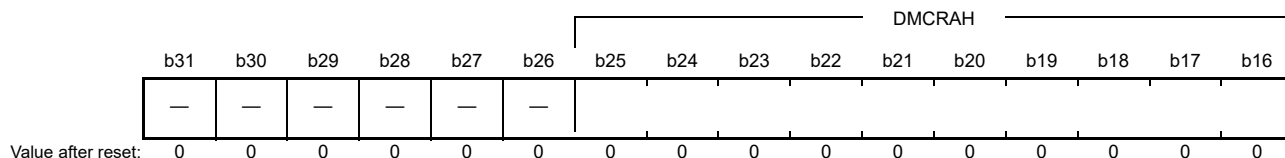
18.2.3 DMA Transfer Count Register (DMCRA)

Address(es): DMAC0.DMCRA 0008 2008h, DMAC1.DMCRA 0008 2048h, DMAC2.DMCRA 0008 2088h, DMAC3.DMCRA 0008 20C8h, DMAC4.DMCRA 0008 2108h, DMAC5.DMCRA 0008 2148h, DMAC6.DMCRA 0008 2188h, DMAC7.DMCRA 0008 21C8h

- Normal transfer mode



- Repeat transfer mode, block transfer mode



Symbol	Bit Name	Description	R/W
DMCRAL	Lower bits of transfer count	Specifies the number of transfer operations	R/W
DMCRAH	Upper bits of transfer count		R/W

Note: Set the same value for the DMCRAH and DMCRAL registers in repeat transfer mode and block transfer mode.

(1) Normal Transfer Mode (DMACm.DMTMD.MD[1:0] Bits = 00b)

The DMCRAL register functions as a 16-bit transfer counter.

The number of transfer operations is one when the setting is 0001h, and 65535 when it is FFFFh. The value is decremented by one each time data is transferred.

When the setting is 0000h, no specific number of transfer operations is set; data transfer is performed with the transfer counter stopped (free running mode).

The DMCRAH register is not used in normal transfer mode. Write 0000h to the DMCRAH register.

(2) Repeat Transfer Mode (DMACm.DMTMD.MD[1:0] Bits = 01b)

The DMCRAH register specifies the repeat size and the DMCRAL register functions as a 10-bit transfer counter. The number of transfer operations is one when the setting is 001h, 1023 when it is 3FFh, and 1024 when it is 000h. In repeat transfer mode, a value in the range of 000h to 3FFh (1 to 1024) can be set for the DMCRAH and DMCRAL registers.

Setting bits 15 to 10 in the DMCRAL register is invalid. Write 0 to these bits.

The value in the DMCRAL register is decremented by one each time data is transferred until it reaches 000h, at which the value in the DMCRAH register is loaded into the DMCRAL register.

(3) Block Transfer Mode (DMACm.DMTMD.MD[1:0] Bits = 10b)

The DMCRAH register specifies the block size and the DMCRAL register functions as a 10-bit block size counter. The block size is one when the setting is 001h, 1023 when it is 3FFh, and 1024 when it is 000h. In block transfer mode, a value in the range of 000h to 3FFh can be set for the DMCRAH and DMCRAL registers.

Setting bits 15 to 10 in the DMCRAL register is invalid. Write 0 to these bits.

The value in the DMCRAL register is decremented by one each time data is transferred until it reaches 000h, at which the value in the DMCRAH register is loaded into the DMCRAL register.

18.2.4 DMA Block Transfer Count Register (DMCRB)

Address(es): DMAC0.DMCRB 0008 200Ch, DMAC1.DMCRB 0008 204Ch, DMAC2.DMCRB 0008 208Ch, DMAC3.DMCRB 0008 20CCh, DMAC4.DMCRB 0008 210Ch, DMAC5.DMCRB 0008 214Ch, DMAC6.DMCRB 0008 218Ch, DMAC7.DMCRB 0008 21CCh



Bit	Description	Setting Range	R/W
b15 to b0	Specifies the block count or repeat count of transfers.	0001h to FFFFh (1 to 65535) 0000h (65536)	R/W

This register specifies the transfer block count in block transfer mode and the repeat count in repeat transfer mode.

In repeat transfer mode, the value is decremented by one when the final data of one repeat size is transferred.

In block transfer mode, the value is decremented by one when the final data of one block size is transferred.

In normal transfer mode, the DMCRB register is not used. The setting is invalid.

18.2.5 DMA Transfer Mode Register (DMTMD)

Address(es): DMAC0.DMTMD 0008 2010h, DMAC1.DMTMD 0008 2050h, DMAC2.DMTMD 0008 2090h, DMAC3.DMTMD 0008 20D0h, DMAC4.DMTMD 0008 2110h, DMAC5.DMTMD 0008 2150h, DMAC6.DMTMD 0008 2190h, DMAC7.DMTMD 0008 21D0h

b15	b14	b13	b12	b11	b10	b9	b8	b7	b6	b5	b4	b3	b2	b1	b0
MD[1:0]		DTS[1:0]		—	—	SZ[1:0]		—	—	—	—	—	—	DCTG[1:0]	
Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0															

Bit	Symbol	Bit Name	Description	R/W
b1, b0	DCTG[1:0]	Transfer Request Source Select	b1 b0 0 0: Software 0 1: Interrupts*1 from peripheral modules or external interrupt input pins 1 0: Setting prohibited 1 1: Setting prohibited	R/W
b7 to b2	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b9, b8	SZ[1:0]	Transfer Data Size Select	b9 b8 0 0: 8 bits 0 1: 16 bits 1 0: 32 bits 1 1: Setting prohibited	R/W
b11, b10	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b13, b12	DTS[1:0]	Repeat Area Select	b13b12 0 0: The destination is specified as the repeat area or block area. 0 1: The source is specified as the repeat area or block area. 1 0: The repeat area or block area is not specified. 1 1: Setting prohibited	R/W
b15, b14	MD[1:0]	Transfer Mode Select	b15b14 0 0: Normal transfer 0 1: Repeat transfer 1 0: Block transfer 1 1: Setting prohibited	R/W

Note 1. DMA request source is selected using the ICU.DMRSRm registers. For details on DMA request sources, refer to Table 15.5, Interrupt Vector Table in section 15, Interrupt Controller (ICUD).

DTS[1:0] Bits (Repeat Area Select)

These bits select either the source or destination as the repeat area in repeat or block transfer mode. In normal transfer mode, setting these bits is invalid.

18.2.6 DMA Interrupt Setting Register (DMINT)

Address(es): DMAC0.DMINT 0008 2013h, DMAC1.DMINT 0008 2053h, DMAC2.DMINT 0008 2093h, DMAC3.DMINT 0008 20D3h, DMAC4.DMINT 0008 2113h, DMAC5.DMINT 0008 2153h, DMAC6.DMINT 0008 2193h, DMAC7.DMINT 0008 21D3h

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	DTIE	ESIE	RPTIE	SARIE	DARIE
Value after reset:	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DARIE	Destination Address Extended Repeat Area Overflow Interrupt Enable	0: Disables an interrupt request for an extended repeat area overflow on the destination address 1: Enables an interrupt request for an extended repeat area overflow on the destination address	R/W
b1	SARIE	Source Address Extended Repeat Area Overflow Interrupt Enable	0: Disables an interrupt request for an extended repeat area overflow on the source address 1: Enables an interrupt request for an extended repeat area overflow on the source address	R/W
b2	RPTIE	Repeat Size End Interrupt Enable	0: Disables the repeat size end interrupt request. 1: Enables the repeat size end interrupt request.	R/W
b3	ESIE	Transfer Escape End Interrupt Enable	0: Disables the transfer escape end interrupt request. 1: Enables the transfer escape end interrupt request.	R/W
b4	DTIE	Transfer End Interrupt Enable	0: Disables the transfer end interrupt request. 1: Enables the transfer end interrupt request.	R/W
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

DARIE Bit (Destination Address Extended Repeat Area Overflow Interrupt Enable)

When an extended repeat area overflow on the destination address occurs while this bit is set to 1, the DMCNT.DTE bit is set to 0. At the same time, the DMSTS.ESIF flag is set to 1 to indicate that an interrupt by an extended repeat area overflow on the destination address is requested.

When block transfer mode is used with the extended repeat area function, an interrupt is requested after completion of a 1-block size transfer. When setting 1 in the DMACm.DMCNT.DTE bit of the channel for which a transfer has been stopped, the transfer is resumed from the state when the transfer is stopped.

When the extended repeat area is not specified for the destination address, this bit is ignored.

SARIE Bit (Source Address Extended Repeat Area Overflow Interrupt Enable)

When an extended repeat area overflow on the source address occurs while this bit is set to 1, the DMCNT.DTE bit is set to 0. At the same time, the DMSTS.ESIF flag is set to 1 to indicate that an interrupt by an extended repeat area overflow on the source address is requested.

When block transfer mode is used with the extended repeat area function, an interrupt is requested after completion of a 1-block size transfer. When setting 1 in the DMACm.DMCNT.DTE bit of the channel for which a transfer has been stopped, the transfer is resumed from the state when the transfer is stopped.

When the extended repeat area is not specified for the source address, this bit is ignored.

RPTIE Bit (Repeat Size End Interrupt Enable)

When this bit is set to 1 in repeat transfer mode, the DMCNT.DTE bit is set to 0 after completion of a 1-repeat size data transfer. At the same time, the DMSTS.ESIF flag is set to 1 to indicate that the repeat size end interrupt request has been generated. The repeat size end interrupt request can be generated even when the DMTMD.DTS[1:0] bits are 10b (= repeat area or block area is not specified).

When this bit is set to 1 in block transfer mode, the DMCNT.DTE bit is set to 0 after completion of a 1-block data transfer in the same way as repeat transfer mode. At the same time, the DMSTS.ESIF flag is set to 1 to indicate that the repeat size end interrupt request has been generated. The repeat size end interrupt request can be generated even when the DMTMD.DTS[1:0] bits are 10b (= repeat area or block area is not specified).

ESIE Bit (Transfer Escape End Interrupt Enable)

This bit enables or disables the transfer escape end interrupt requests (repeat size end interrupt request and extended repeat area overflow interrupt request) that are generated during DMA transfer.

The transfer escape end interrupt is generated when the DMSTS.ESIF flag is set to 1 with this bit set to 1. The transfer escape end interrupt is cleared by setting this bit or the DMSTS.ESIF flag to 0.

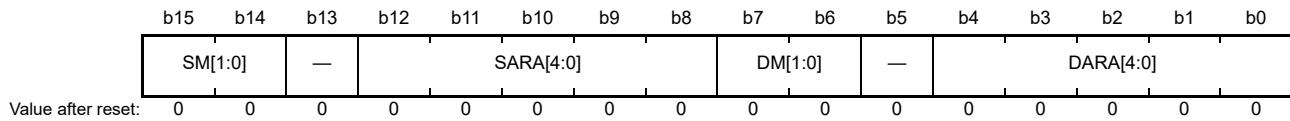
DTIE Bit (Transfer End Interrupt Enable)

This bit enables or disables the transfer end interrupt request to be generated on completion of a specified number of data transfers.

The transfer end interrupt is generated when the DMSTS.DTIF bit is set to 1 with this bit set to 1. The transfer end interrupt is cleared by setting this bit or the DMSTS.DTIF bit to 0.

18.2.7 DMA Address Mode Register (DMAMD)

Address(es): DMAC0.DMAMD 0008 2014h, DMAC1.DMAMD 0008 2054h, DMAC2.DMAMD 0008 2094h, DMAC3.DMAMD 0008 20D4h, DMAC4.DMAMD 0008 2114h, DMAC5.DMAMD 0008 2154h, DMAC6.DMAMD 0008 2194h, DMAC7.DMAMD 0008 21D4h



Bit	Symbol	Bit Name	Description	R/W
b4 to b0	DARA[4:0]	Destination Address Extended Repeat Area	Specifies the extended repeat area on the destination address. For details on the settings, refer to Table 18.2.	R/W
b5	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b7, b6	DM[1:0]	Destination Address Update Mode	b7 b6 0 0: Destination address is fixed. 0 1: Offset addition*1 1 0: Destination address is incremented. 1 1: Destination address is decremented.	R/W
b12 to b8	SARA[4:0]	Source Address Extended Repeat Area	Specifies the extended repeat area on the source address. For details on the settings, refer to Table 18.2.	R/W
b13	—	Reserved	This bit is read as 0. The write value should be 0.	R/W
b15, b14	SM[1:0]	Source Address Update Mode	b15b14 0 0: Source address is fixed. 0 1: Offset addition*1 1 0: Source address is incremented. 1 1: Source address is decremented.	R/W

Note 1. Offset addition can be specified only for DMAC0.

DARA[4:0] Bits (Destination Address Extended Repeat Area)

These bits specify the extended repeat area on the destination address. The extended repeat area function is realized by updating the specified lower address bits with the remaining upper address bits fixed. The size of the extended repeat area can be any power of two between 2 bytes and 128 Mbytes. That is, the interval between settings is any power of two. When the lower address overflows the extended repeat area by address increment, the start address of the extended repeat area is set. Similarly, when the lower address underflows the extended repeat area by address decrement, the end address of the extended repeat area is set.

When the repeat area or block area is specified as a transfer destination, do not specify the extended repeat area on the destination address. When repeat transfer or block transfer is selected, and when DMACm.DMTMD.DTS[1:0] = 00b (the transfer destination is specified as the repeat area or block area), write 00000b in the DARA[4:0] bits.

An interrupt can be requested when an overflow or underflow occurs in the extended repeat area with the DMINT.DARIE bit set to 1. Table 18.2 lists the settings and the corresponding extended repeat areas.

DM[1:0] Bits (Destination Address Update Mode)

These bits select the mode of updating the destination address.

When increment is selected and the DMTMD.SZ[1:0] bits are set to 00b, 01b, and 10b, the destination address is incremented by 1, 2, and 4, respectively.

When decrement is selected and the DMTMD.SZ[1:0] bits are set to 00b, 01b, and 10b, the destination address is decremented by 1, 2, and 4, respectively.

When offset addition is selected, the offset specified by the DMAC0.DMOFR register is added to the address.

Offset addition can be specified only for DMAC0.

SARA[4:0] Bits (Source Address Extended Repeat Area)

These bits specify the extended repeat area on the source address. The extended repeat area function is realized by updating the specified lower address bits with the remaining upper address bits fixed. The size of the extended repeat area can be any power of two between 2 bytes and 128 Mbytes.

When the lower address overflows the extended repeat area by address increment, the start address of the extended repeat area is set. Similarly, when the lower address underflows the extended repeat area by address decrement, the end address of the extended repeat area is set.

When the repeat area or block area is specified as a transfer source, do not specify the extended repeat area on the source address. When repeat transfer or block transfer is selected, and when DMACm.DMTMD.DTS[1:0] = 01b (the transfer source is specified as the repeat area or block area), write 00000b in the SARA[4:0] bits.

An interrupt can be requested when an overflow or underflow occurs in the extended repeat area with the DMINT.SARIE bit set to 1. Table 18.2 lists the settings and the corresponding extended repeat areas.

SM[1:0] Bits (Source Address Update Mode)

These bits select the mode of updating the source address.

When increment is selected and the DMTMD.SZ[1:0] bits are set to 00b, 01b, and 10b, the source address is incremented by 1, 2, and 4, respectively.

When decrement is selected and the DMTMD.SZ[1:0] bits are set to 00b, 01b, and 10b, the source address is decremented by 1, 2, and 4, respectively.

When offset addition is selected, the offset specified by the DMAC0.DMOFR register is added to the address.

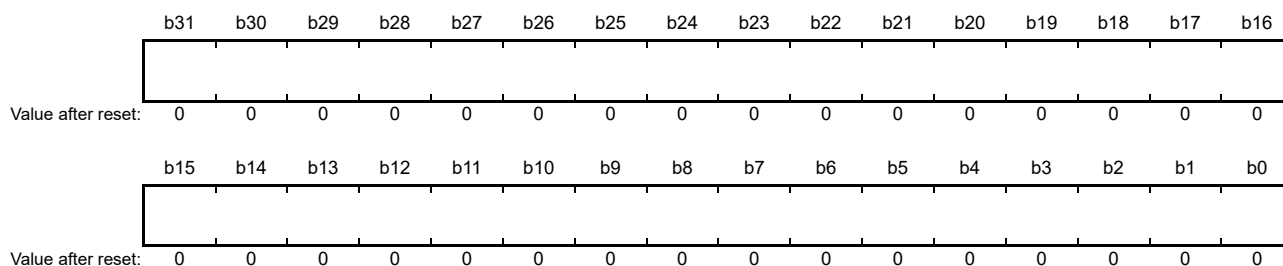
Offset addition can be specified only for DMAC0.

Table 18.2 SARA[4:0] or DARA[4:0] Settings and Corresponding Repeat Areas

SARA[4:0] or DARA[4:0]	Extended Repeat Area
00000b	Not specified
00001b	2 bytes specified as extended repeat area by the lower 1 bit of the address
00010b	4 bytes specified as extended repeat area by the lower 2 bits of the address
00011b	8 bytes specified as extended repeat area by the lower 3 bits of the address
00100b	16 bytes specified as extended repeat area by the lower 4 bits of the address
00101b	32 bytes specified as extended repeat area by the lower 5 bits of the address
00110b	64 bytes specified as extended repeat area by the lower 6 bits of the address
00111b	128 bytes specified as extended repeat area by the lower 7 bits of the address
01000b	256 bytes specified as extended repeat area by the lower 8 bits of the address
01001b	512 bytes specified as extended repeat area by the lower 9 bits of the address
01010b	1 Kbyte specified as extended repeat area by the lower 10 bits of the address
01011b	2 Kbytes specified as extended repeat area by the lower 11 bits of the address
01100b	4 Kbytes specified as extended repeat area by the lower 12 bits of the address
01101b	8 Kbytes specified as extended repeat area by the lower 13 bits of the address
01110b	16 Kbytes specified as extended repeat area by the lower 14 bits of the address
01111b	32 Kbytes specified as extended repeat area by the lower 15 bits of the address
10000b	64 Kbytes specified as extended repeat area by the lower 16 bits of the address
10001b	128 Kbytes specified as extended repeat area by the lower 17 bits of the address
10010b	256 Kbytes specified as extended repeat area by the lower 18 bits of the address
10011b	512 Kbytes specified as extended repeat area by the lower 19 bits of the address
10100b	1 Mbyte specified as extended repeat area by the lower 20 bits of the address
10101b	2 Mbytes specified as extended repeat area by the lower 21 bits of the address
10110b	4 Mbytes specified as extended repeat area by the lower 22 bits of the address
10111b	8 Mbytes specified as extended repeat area by the lower 23 bits of the address
11000b	16 Mbytes specified as extended repeat area by the lower 24 bits of the address
11001b	32 Mbytes specified as extended repeat area by the lower 25 bits of the address
11010b	64 Mbytes specified as extended repeat area by the lower 26 bits of the address
11011b	128 Mbytes specified as extended repeat area by the lower 27 bits of the address
11100b to 11111b	Setting prohibited.

18.2.8 DMA Offset Register (DMOFR)

Address(es): DMAC0.DMOFR 0008 2018h

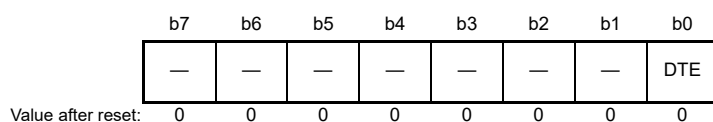


Bit	Description	Setting Range	R/W
b31 to b0	Specifies the offset when offset addition is selected as the address update mode for transfer source or destination.	0000 0000h to 00FF FFFFh (0 bytes to (16 M – 1) bytes) FF00 0000h to FFFF FFFFh (–16 Mbytes to –1 byte)	R/W

Write to this register while the DMAC operation is stopped or DMA transfer is disabled (not during data transfer).
Setting bits 31 to 25 is invalid; a value of bit 24 is extended to bits 31 to 25. Reading DMOFR returns the extended value.

18.2.9 DMA Transfer Enable Register (DMCNT)

Address(es): DMAC0.DMCNT 0008 201Ch, DMAC1.DMCNT 0008 205Ch, DMAC2.DMCNT 0008 209Ch, DMAC3.DMCNT 0008 20DCh, DMAC4.DMCNT 0008 211Ch, DMAC5.DMCNT 0008 215Ch, DMAC6.DMCNT 0008 219Ch, DMAC7.DMCNT 0008 21DCh



Bit	Symbol	Bit Name	Description	R/W
b0	DTE	DMA Transfer Enable	0: Disables DMA transfer. 1: Enables DMA transfer.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

DTE Bit (DMA Transfer Enable)

When the DMAST.DMST bit is set to 1 (DMAC module start) and this bit is set to 1 (DMA transfer is enabled), DMA transfer can be started for the corresponding channel.

[Setting condition]

- When 1 is written to this bit.

[Clearing conditions]

- When 0 is written to this bit.
- When the specified total volume of data transfer is completed.
- When DMA transfer is stopped by the repeat size end interrupt.
- When DMA transfer is stopped by the extended repeat area overflow interrupt.

18.2.10 DMA Software Start Register (DMREQ)

Address(es): DMAC0.DMREQ 0008 201Dh, DMAC1.DMREQ 0008 205Dh, DMAC2.DMREQ 0008 209Dh, DMAC3.DMREQ 0008 20DDh, DMAC4.DMREQ 0008 211Dh, DMAC5.DMREQ 0008 215Dh, DMAC6.DMREQ 0008 219Dh, DMAC7.DMREQ 0008 21DDh

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	CLRS	—	—	—	SWREQ
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	SWREQ	DMA Software Start	0: DMA transfer is not requested. 1: DMA transfer is requested.	R/W
b3 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W
b4	CLRS	DMA Software Start Bit Auto Clear Select	0: SWREQ bit is cleared after DMA transfer is started by software. 1: SWREQ bit is not cleared after DMA transfer is started by software.	R/W
b7 to b5	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

SWREQ Bit (DMA Software Start)

When 1 is written to this bit, a DMA transfer request is generated. After DMA transfer is started in response to the request, this bit is set to 0 if the CLRS bit is set to 0. This bit is not set to 0 while the CLRS bit is set to 1. In this case, a DMA transfer request can be issued again after completion of a transfer.

Note that, however, setting this bit is valid and DMA transfer by software is enabled only when the DMTMD.DCTG[1:0] bits are set to 00b (DMA request source is software).

Setting this bit is invalid when the DMTMD.DCTG[1:0] bits are set to a value other than 00b.

To start DMA transfer by software with the CLRS bit being 0, ensure that the SWREQ bit is 0, and then write 1 to the SWREQ bit.

[Setting condition]

- When 1 is written to this bit.

[Clearing conditions]

- When a DMA transfer request by software is accepted and DMA transfer is started while the CLRS bit is set to 0 (the SWREQ bit is cleared after DMA transfer is started by software).
- When 0 is written to this bit.

CLRS Bit (DMA Software Start Bit Auto Clear Select)

This bit specifies whether to clear the SWREQ bit to 0 after DMA transfer is started in response to the DMA transfer request generated by setting the SWREQ bit to 1. With this bit set to 0, the SWREQ bit is set to 0 after DMA transfer is started. With this bit set to 1, the SWREQ bit is not set to 0. In this case, a DMA transfer request can be issued again after completion of a transfer.

18.2.11 DMA Status Register (DMSTS)

Address(es): DMAC0.DMSTS 0008 201Eh, DMAC1.DMSTS 0008 205Eh, DMAC2.DMSTS 0008 209Eh, DMAC3.DMSTS 0008 20DEh, DMAC4.DMSTS 0008 211Eh, DMAC5.DMSTS 0008 215Eh, DMAC6.DMSTS 0008 219Eh, DMAC7.DMSTS 0008 21DEh

b7	b6	b5	b4	b3	b2	b1	b0
ACT	—	—	DTIF	—	—	—	ESIF

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Bit Name	Description	R/W
b0	ESIF	Transfer Escape End Interrupt Flag	0: A transfer escape end interrupt has not been generated. 1: A transfer escape end interrupt has been generated.	R/W*1
b3 to b1	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R
b4	DTIF	Transfer End Interrupt Flag	0: A transfer end interrupt has not been generated. 1: A transfer end interrupt has been generated.	R/W*1
b6, b5	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R
b7	ACT	DMA Active Flag	0: DMAC operation is suspended. 1: DMAC is operating.	R

Note 1. Only 0 can be written to clear the flag.

ESIF Flag (Transfer Escape End Interrupt Flag)

This flag indicates that the transfer escape end interrupt has been generated.

[Setting conditions]

- When 1-repeat size data transfer is completed in repeat transfer mode with the DMINT.RPTIE bit set to 1.
- When 1-block data transfer is completed in block transfer mode with the DMINT.RPTIE bit set to 1.
- When an extended repeat area overflow on the source address occurs while the DMINT.SARIE bit is set to 1 and the DMAMD.SARA[4:0] bits are set to a value other than 00000b (extended repeat area is specified on the transfer source address)
- When an extended repeat area overflow on the destination address occurs while the DMINT.DARIE bit is set to 1 and the DMAMD.DARA[4:0] bits are set to a value other than 00000b (extended repeat area is specified on the transfer destination address)

[Clearing conditions]

- When 0 is written to this bit.
- When 1 is written to the DMCNT.DTE bit.

DTIF Flag (Transfer End Interrupt Flag)

This flag indicates that the transfer end interrupt has been generated.

[Setting conditions]

- When the specified number of unit-transfers are completed in normal transfer mode (the value of the DMCRAL register becoming 0 on completion of transfer)
- When the specified number of repeat transfer operations are completed in repeat transfer mode (the value of the DMCRB register becoming 0 on completion of transfer)
- When the specified number of blocks have been transferred in block transfer mode (the value of the DMCRB register becoming 0 on completion of transfer)

[Clearing conditions]

- When 0 is written to this bit
- When 1 is written to the DMCNT.DTE bit

ACT Flag (DMA Active Flag)

This flag indicates whether the DMAC is in the idle or active state.

[Setting condition]

- When the DMAC starts data transfer operation

[Clearing condition]

- When data transfer in response to one transfer request is completed

18.2.12 DMA Request Source Flag Control Register (DMCSL)

Address(es): DMAC0.DMCSL 0008 201Fh, DMAC1.DMCSL 0008 205Fh, DMAC2.DMCSL 0008 209Fh, DMAC3.DMCSL 0008 20DFh, DMAC4.DMCSL 0008 211Fh, DMAC5.DMCSL 0008 215Fh, DMAC6.DMCSL 0008 219Fh, DMAC7.DMCSL 0008 21DFh

b7	b6	b5	b4	b3	b2	b1	b0
—	—	—	—	—	—	—	DISEL
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b0	DISEL	Interrupt Select	0: At the beginning of transfer, clear the interrupt status flag of the request source to 0. 1: At the end of transfer, the interrupt status flag of the request source issues an interrupt to the CPU.	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

DISEL Bit (Interrupt Select)

This bit selects whether the interrupt status flag of the request source is set to 0 or issues an interrupt request to the CPU, at the beginning of DMA transfer.

When DMTMD.DCTG[1:0] = 00b (trigger by software), the setting of the DISEL bit does not affect the operation.

18.2.13 DMAC Module Start Register (DMAST)

Address(es): 0008 2200h

	b7	b6	b5	b4	b3	b2	b1	b0
	—	—	—	—	—	—	—	DMST
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Bit Name	Description	R/W
b0	DMST	DMAC Module Start	0: DMAC module stop 1: DMAC module start	R/W
b7 to b1	—	Reserved	These bits are read as 0. The write value should be 0.	R/W

DMST Bit (DMAC Module Start)

When this bit is set to 1, DMAC is ready to accept transfer requests for all channels.

When 1 is written to the DMACm.DMCNT.DTE bit (DMA transfer is enabled) of multiple channels and then this bit is set to 1 (DMAC module start), the corresponding multiple channels can be placed in the transfer request acceptable state at the same time.

When the DMST bit is set to 0 during DMA transfer, DMA transfer is suspended after completion of the current data transfer corresponding to a single transfer request. DMA transfer can be resumed by setting the DMST bit to 1 again.

[Setting condition]

- When 1 is written to this bit

[Clearing condition]

- When 0 is written to this bit

18.2.14 DMAC74 Interrupt Status Monitor Register (DMIST)

Address(es): 0008 2204h

b7	b6	b5	b4	b3	b2	b1	b0
DMIS7	DMIS6	DMIS5	DMIS4	—	—	—	—
0	0	0	0	0	0	0	0

Value after reset:

Bit	Symbol	Bit Name	Description	R/W
b3 to b0	—	Reserved	These bits are read as 0. Writing to these bits has no effect.	R
b4	DMIS4	DMAC4 Interrupt Status Flag	0: DMAC4 interrupt is not requested. 1: DMAC4 interrupt is requested.	R
b5	DMIS5	DMAC5 Interrupt Status Flag	0: DMAC5 interrupt is not requested. 1: DMAC5 interrupt is requested.	R
b6	DMIS6	DMAC6 Interrupt Status Flag	0: DMAC6 interrupt is not requested. 1: DMAC6 interrupt is requested.	R
b7	DMIS7	DMAC7 Interrupt Status Flag	0: DMAC7 interrupt is not requested. 1: DMAC7 interrupt is requested.	R

DMISm Flag (DMACm Interrupt Status Flag) (m = 4 to 7)

This flag monitors the DMACm interrupt request. Writing to this flag will be ignored.

While the DMACm.DMINT.DTIE bit is 1 and the DMACm.DMSTS.DTIF flag is 1, or the DMACm.DMINT.ESIE bit is 1 and the DMACm.DMSTS.ESIF flag is 1, the DMIST.DMISm flag is set to 1.

18.3 Operation

18.3.1 Transfer Mode

(1) Normal Transfer Mode

In normal transfer mode, one data is transferred by one transfer request. A maximum of 65535 can be set as the number of transfer operations using the DMACm.DMCRAL register. When these bits are set to 0000h, no specific number of transfer operations is set; data transfer is performed with the transfer counter stopped (free running mode). Setting the DMACm.DMCRB register is invalid in normal transfer mode. Except in free running mode, a transfer end interrupt request can be generated after completion of the specified number of transfer operations.

Table 18.3 summarizes the register update operation in normal transfer mode, and Figure 18.2 shows the operation in normal transfer mode.

Table 18.3 Register Update Operation in Normal Transfer Mode

Register	Function	Update Operation after Completion of a Transfer by One Transfer Request
DMACm.DMSAR	Transfer source address	Increment/decrement/fixed/offset addition*1
DMACm.DMDAR	Transfer destination address	Increment/decrement/fixed/offset addition*1
DMACm.DMCRAL	Transfer counter	Decrement by one/not updated (in free running mode)
DMACm.DMCRAH	—	Not updated (Not used in normal transfer mode)
DMACm.DMCRB	—	Not updated (Not used in normal transfer mode)

Note 1. Offset addition can be specified only for DMAC0.

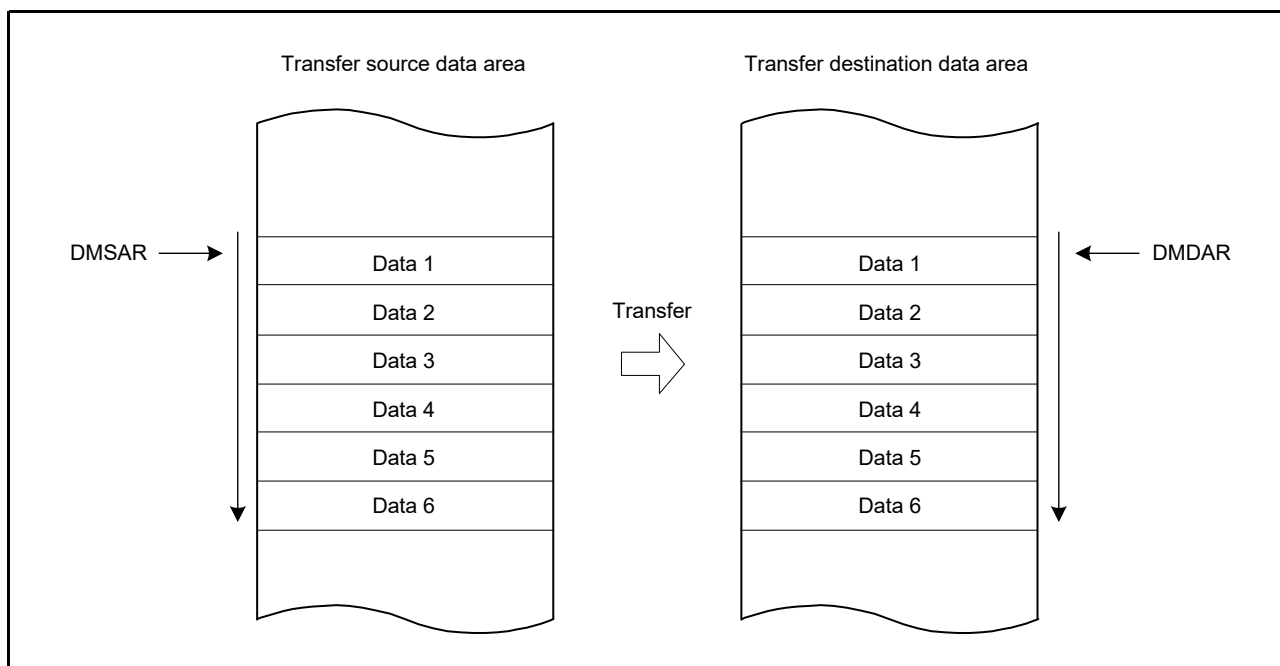


Figure 18.2 Operation in Normal Transfer Mode

(2) Repeat Transfer Mode

In repeat transfer mode, one data is transferred by one transfer request.

A maximum of 1K data can be set as a total repeat transfer size using the DMACm.DMCRA register.

A maximum of 64K can be set as the number of repeat transfer operations using the DMACm.DMCRB register; therefore, a maximum of 64M data (1K data × 64K counts of repeat transfer operations) can be set as a total data transfer size.

Either the transfer source or transfer destination can be specified as a repeat area. When transfer of the repeat size data is completed, the address of the specified repeat area (DMSAR or DMDAR of the DMACm) returns to the transfer start address. When data of the specified repeat size has all been transferred in repeat transfer mode, DMA transfer can be stopped and the repeat size end interrupt can be requested. DMA transfer can be resumed by writing 1 to the DMACm.DMCNT.DTE bit in the repeat size end interrupt handling.

A transfer end interrupt request can be generated after completion of the specified number of repeat transfer operations. Table 18.4 summarizes the register update operation in repeat transfer mode, and Figure 18.3 shows the operation in repeat transfer mode.

Table 18.4 Register Update Operation in Repeat Transfer Mode

Register	Function	Update Operation after Completion of a Transfer by One Transfer Request	
		When the DMACm.DMCRAL register is not 1	When the DMACm.DMCRAL register is 1 (Transfer of the Last Data in Repeat Size)
DMACm.DMSAR	Transfer source address	Increment/decrement/fixed/offset addition*1	- DMACm.DMTMD.DTS[1:0] = 00b Increment/decrement/fixed/offset addition*1 - DMACm.DMTMD.DTS[1:0] = 01b Initial value of DMACm.DMSAR - DMACm.DMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition*1
DMACm.DMDAR	Transfer destination address	Increment/decrement/fixed/offset addition*1	- DMACm.DMTMD.DTS[1:0] = 00b Initial value of DMACm.DMDAR - DMACm.DMTMD.DTS[1:0] = 01b Increment/decrement/fixed/offset addition*1 - DMACm.DMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition*1
DMACm.DMCRAH	Repeat size	Not updated	Not updated
DMACm.DMCRAL	Transfer counter	Decrement by one	DMACm.DMCRAH
DMACm.DMCRB	Repeat-count counter	Not updated	Decrement by one

Note 1. Offset addition can be specified only for DMAC0.

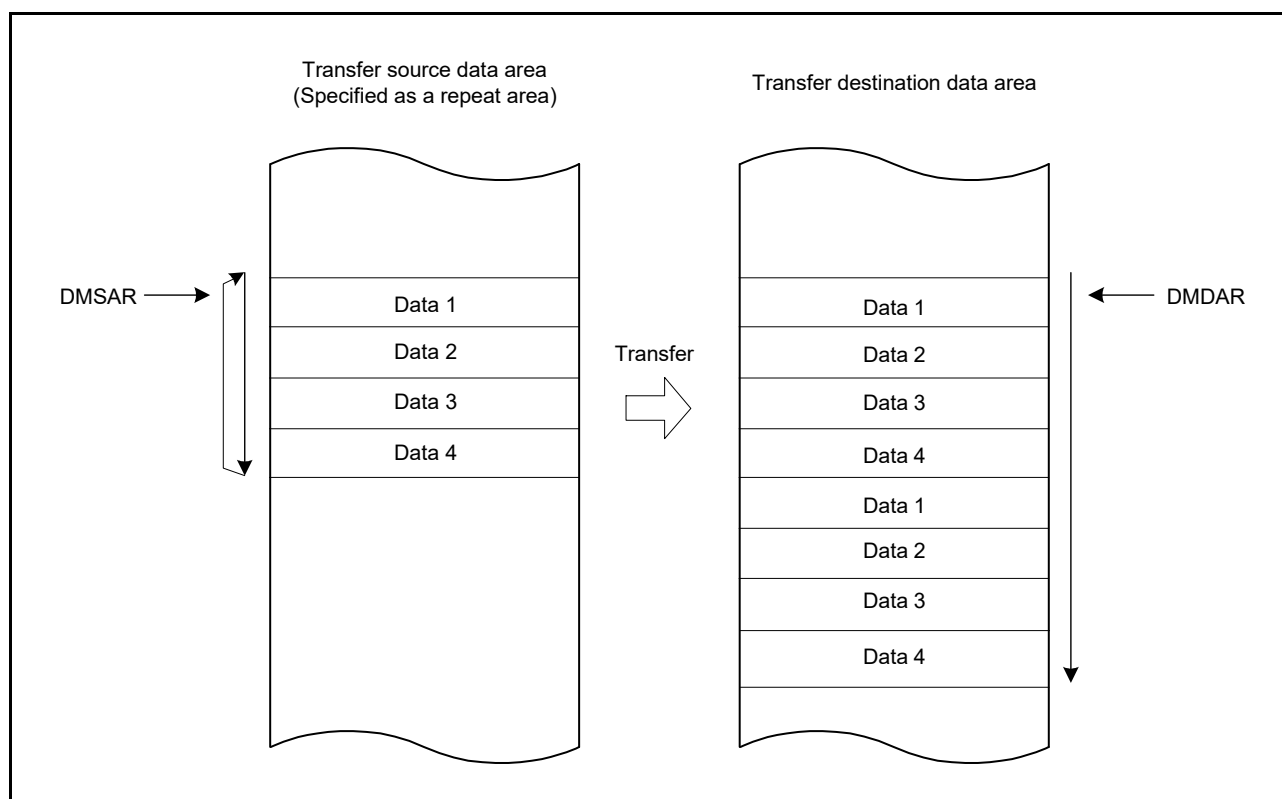


Figure 18.3 Operation in Repeat Transfer Mode

(3) Block Transfer Mode

In block transfer mode, a single block data is transferred by one transfer request.

A maximum of 1K data can be set as a total block transfer size using the DMACm.DMCRA register.

A maximum of 64K can be set as the number of block transfer operations using the DMACm.DMCRB register; therefore, a maximum of 64M data (1K data × 64K counts of block transfer operations) can be set as a total data transfer size.

Either the transfer source or transfer destination can be specified as a block area. When transfer of a single block data is completed, the address of the specified block area (DMSAR or DMDAR of the DMACm) returns to the transfer start address. When a single block data has all been transferred in block transfer mode, DMA transfer can be stopped and the repeat size end interrupt can be requested. DMA transfer can be resumed by writing 1 to the DMACm.DMCNT.DTE bit in the repeat size end interrupt handling.

Transfer end interrupt request can be generated after completion of the specified number of block transfer operations.

Table 18.5 summarizes the register update operation in block transfer mode, and Figure 18.4 shows the operation in block transfer mode.

Table 18.5 Register Update Operation in Block Transfer Mode

Register	Function	Update Operation after Completion of Single-Block Transfer by One Transfer Request
DMACm.DMSAR	Transfer source address	- DMACm.DMTMD.DTS[1:0] = 00b Increment/decrement/fixed/offset addition*1 - DMACm.DMTMD.DTS[1:0] = 01b Initial value of DMACm.DMSAR - DMACm.DMTMD.DTS[1:0] = 10b Increment/decrement/fixed/offset addition*1
DMACm.DMDAR	Transfer destination address	- DMACm.DMTMD.DTS[1:0] = 00 b Initial value of DMACm.DMDAR - DMACm.DMTMD.DTS[1:0] = 01 b Increment/decrement/fixed/offset addition*1 - DMACm.DMTMD.DTS[1:0] = 10 b Increment/decrement/fixed/offset addition*1
DMACm.DMCRAH	Block size	Not updated
DMACm.DMCRAL	Block size counter	DMACm.DMCRAH
DMACm.DMCRB	Transfer-block counter	Decrement by one

Note 1. Offset addition can be specified only for DMAC0.

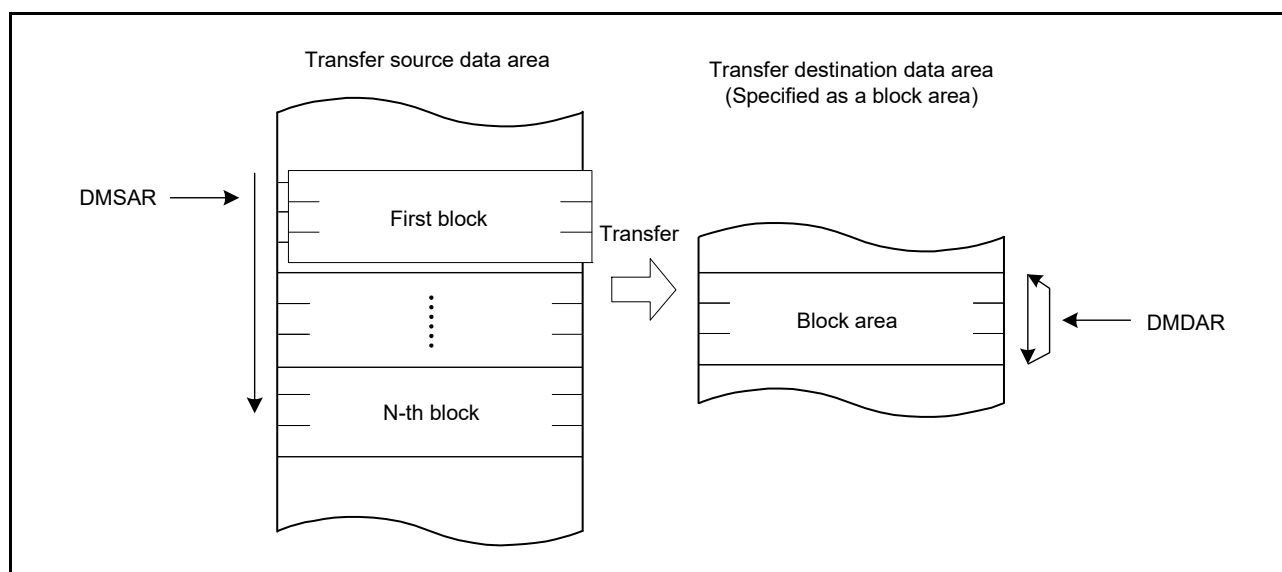


Figure 18.4 Operation in Block Transfer Mode

18.3.2 Extended Repeat Area Function

The DMAC supports a function to specify the extended repeat areas on the transfer source and destination addresses. With the extended repeat areas set, the address registers repeatedly indicate the addresses of the specified extended repeat areas.

The extended repeat areas can be specified separately to the transfer source address register (DMSAR) and transfer destination address register (DMDAR) of DMACm.

The extended repeat area on the source address is specified by the DMACmDMAMD.SARA[4:0] bits. The extended repeat area on the destination address is specified by the DMACmDMAMD.DARA[4:0] bits. The size can be specified separately for the source and destination sides.

However, the area (of transfer source or transfer destination) which is specified as the repeat area or block area should not be specified as the extended repeat area.

When the address register value reaches the end address of the extended repeat area and the extended repeat area overflows, DMA transfer is stopped and an interrupt by an extended repeat area overflow can be requested. When an overflow occurs in the extended repeat area on the transfer source while the DMACm.DMINT.SARIE bit is set to 1, the DMACm.DMSTS.ESIF flag is set to 1 and the DMACm.DMCNT.DTE bit is set to 0 to stop DMA transfer. At this time, if the DMACm.DMINT.ESIE bit is set to 1, an interrupt by an extended repeat area overflow is requested. When the DMACm.DMINT.DARIE bit is set to 1, the destination address register becomes a target to apply the function. DMA transfer can be resumed by writing 1 to the DMACm.DMCNT.DTE bit in the interrupt handling.

Figure 18.5 shows an example of the extended repeat area operation.

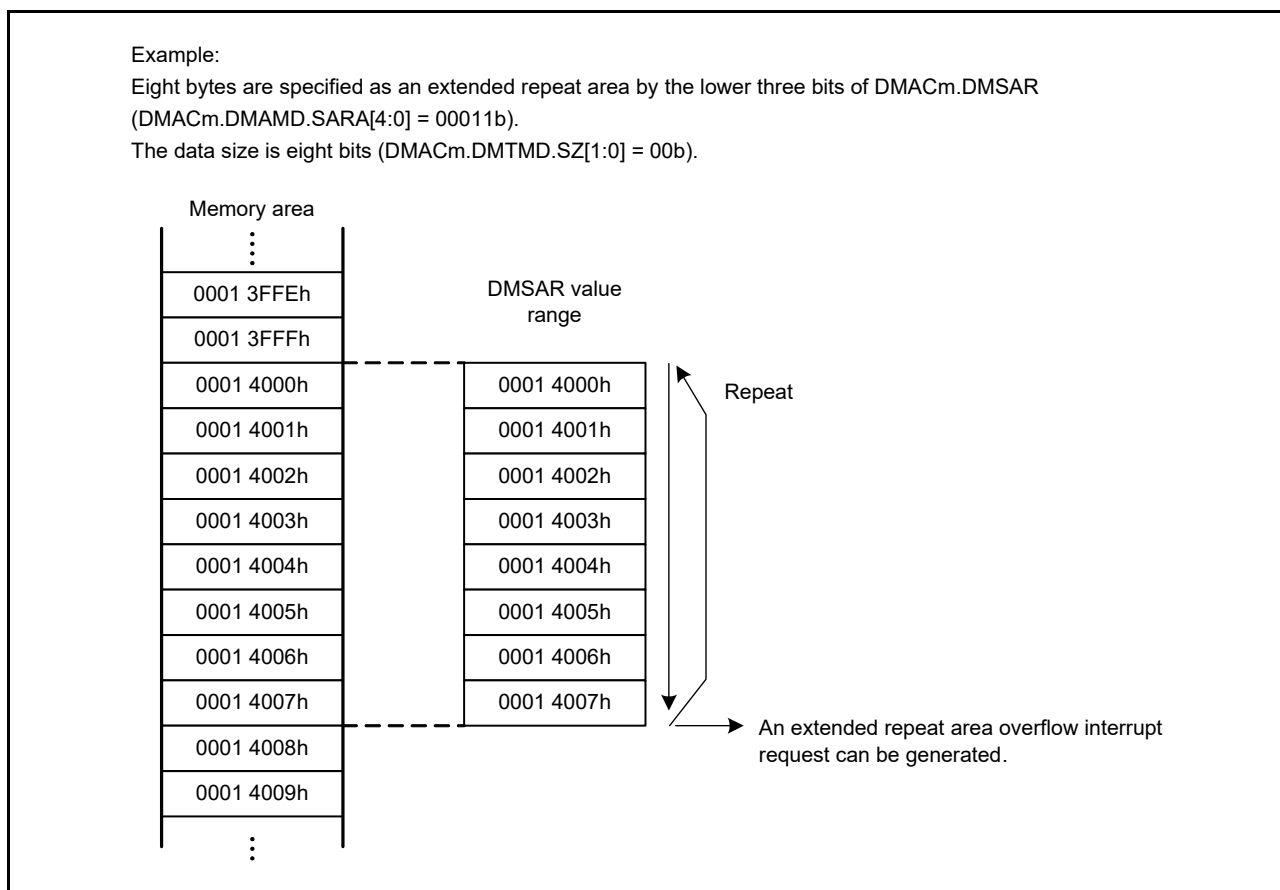


Figure 18.5 Example of Extended Repeat Area Operation

When an interrupt by an extended repeat area overflow is used in block transfer mode, the following should be taken into consideration.

When a transfer is stopped by an interrupt by an extended repeat area overflow, the address register must be set so that the block size is a power of 2 or the block size boundary is aligned with the extended repeat area boundary. When an overflow on the extended repeat area occurs during a transfer of one block, the interrupt by the overflow is suspended until transfer of the block is completed, and the transfer overruns.

Figure 18.6 shows an example when the extended repeat area function is used in block transfer mode.

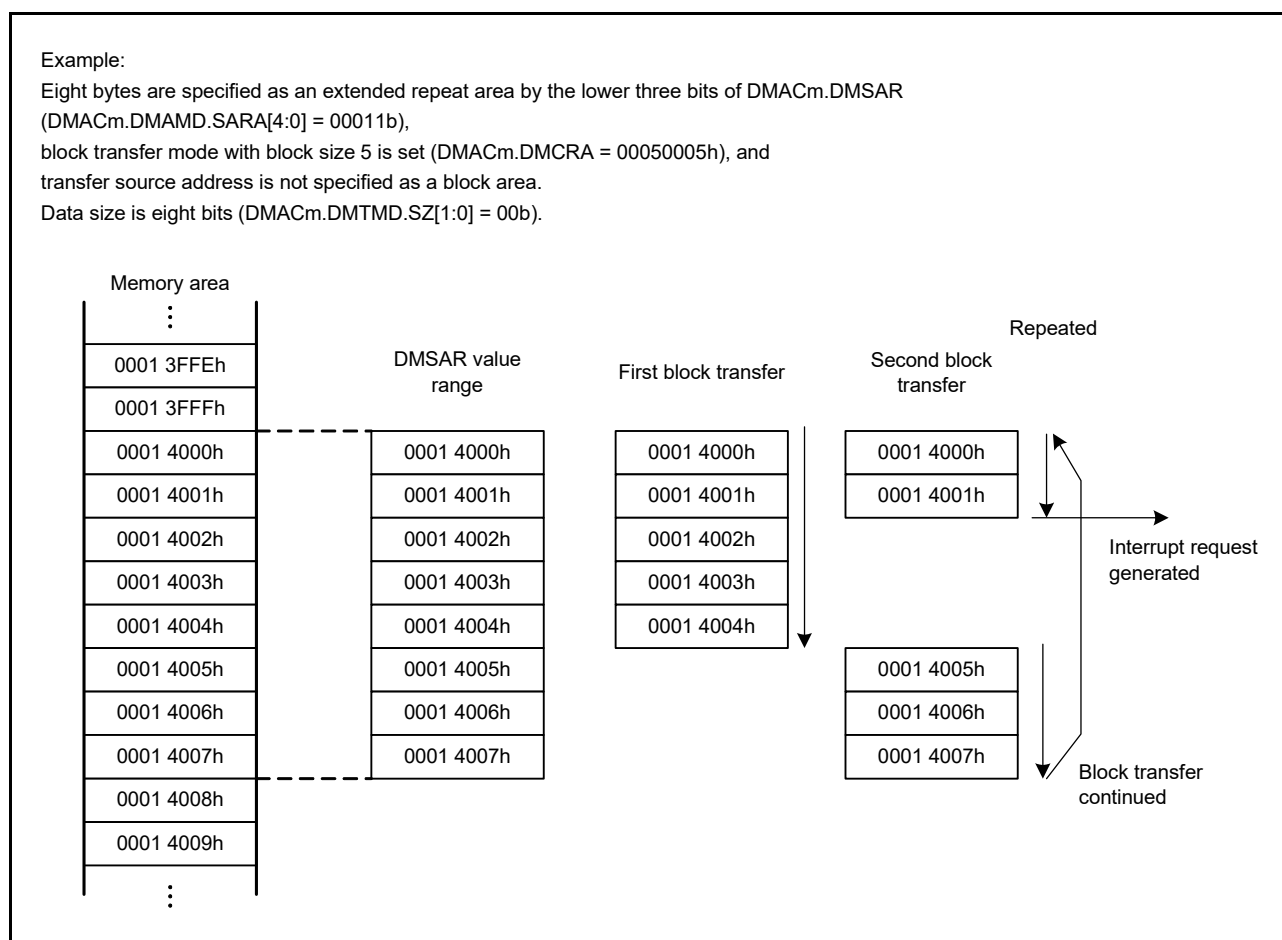


Figure 18.6 Example of Extended Repeat Area Function in Block Transfer Mode

18.3.3 Address Update Function using Offset

The source and destination addresses can be updated by fixing, increment, decrement, or offset addition. When the offset addition is selected, the offset specified by the DMA offset register (DMAC0.DMOFR) is added to the address every time the DMAC performs one data transfer. This function realizes a data transfer where addresses are allocated to separated areas.

Offset subtraction can also be realized by setting a negative value in the DMAC0.DMOFR register. In this case, the negative value must be 2's complement.

Address update function using offset can be specified only for the DMAC0 channel.

Table 18.6 lists the address update method in each address update mode.

Table 18.6 Address Update Method in Each Address Update Mode

Address Update Mode	Settings of DMACm.DMAMD.SM[1:0] and DMACm.DMAMD.DM[1:0] for Address Update Modes	Address Update Method (for Different DMACm.DMTMDSZ[1:0] Settings)		
		SZ[1:0] = 00b	SZ[1:0] = 01b	SZ[1:0] = 10b
Address fixed	00b	Fixed		
Offset addition	01b	+DMACm.DMOFR*1		
Increment	10b	+1	+2	+4
Decrement	11b	-1	-2	-4

Note 1. When setting a negative value in the DMA offset register, the value must be 2's complement. The 2's complement is obtained by the following formula.

2's complement of a negative offset value = $\sim(\text{offset}) + 1$ ($\sim$: bit inversion)

(1) Basic Transfer Using Offset Addition

Figure 18.7 shows an example of address updating using offset addition.

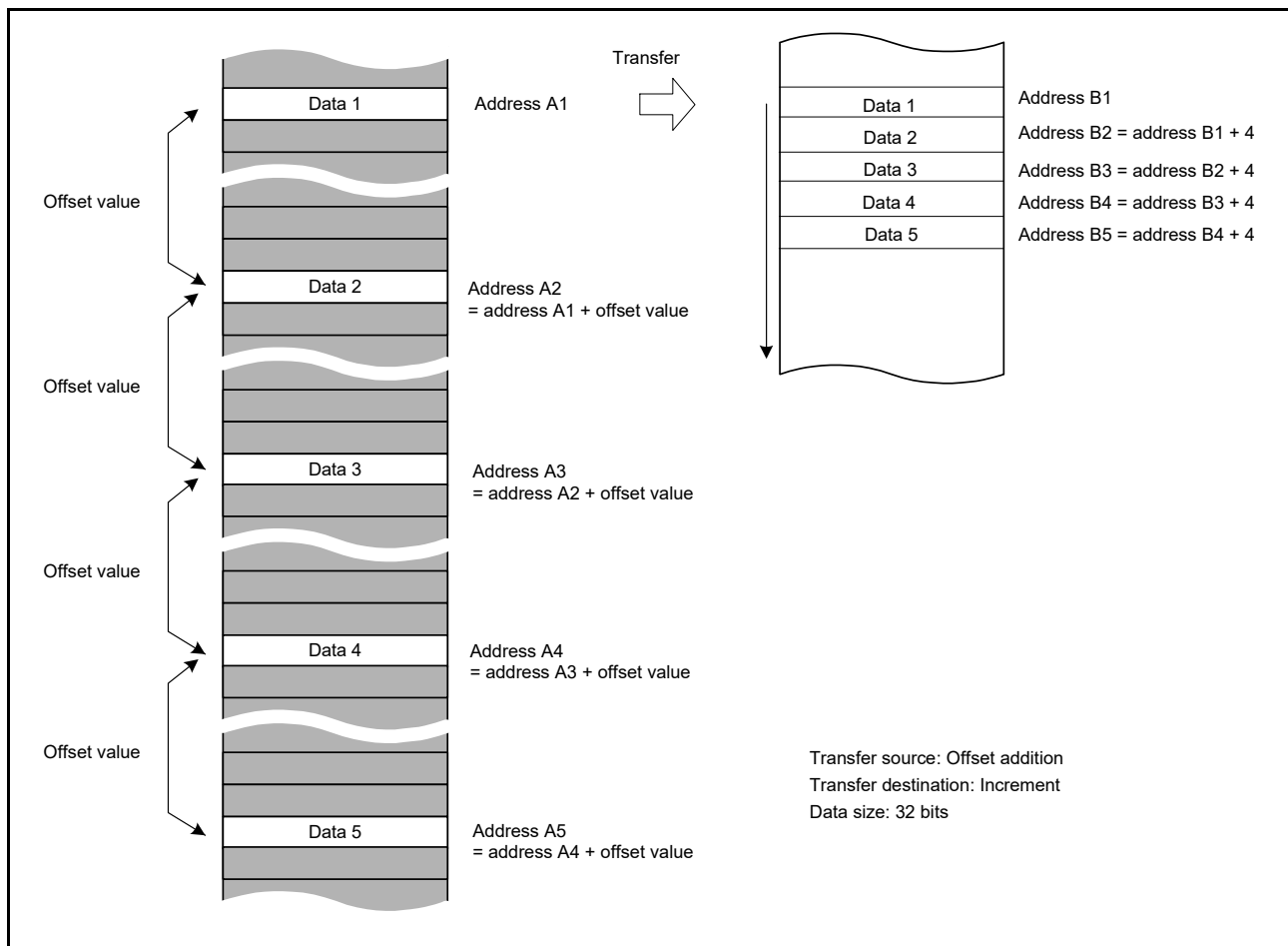


Figure 18.7 Example of Address Updating by Offset Addition

In Figure 18.7, the transfer data is 32 bits long, and offset addition and increment are set as the transfer source address update mode and transfer destination address update mode, respectively. The second and subsequent data is each read from the transfer source address obtained by adding the offset value to the previous address. The data read from the addresses at the specified intervals is written to the continuous locations on the destination.

(2) Example of XY Conversion Using Offset Addition

Figure 18.8 shows the XY conversion using offset addition in repeat transfer mode.

Settings are as follows:

- DMAC0.DMAMD register: Transfer source address update mode: Offset addition
- DMAC0.DMAMD register: Transfer destination address update mode: Destination address is incremented.
- DMAC0.DMTMD register: Transfer data size select: 32 bits
- DMAC0.DMTMD register: Transfer mode select: Repeat transfer
- DMAC0.DMTMD register: Repeat area select: The source is specified as the repeat area.
- DMAC0.DMOFR register: Offset address: 10h
- DMAC0.DMCRA register: Repeat size: 4h
- DMAC0.DMINT register: The repeat size end interrupt is enabled.

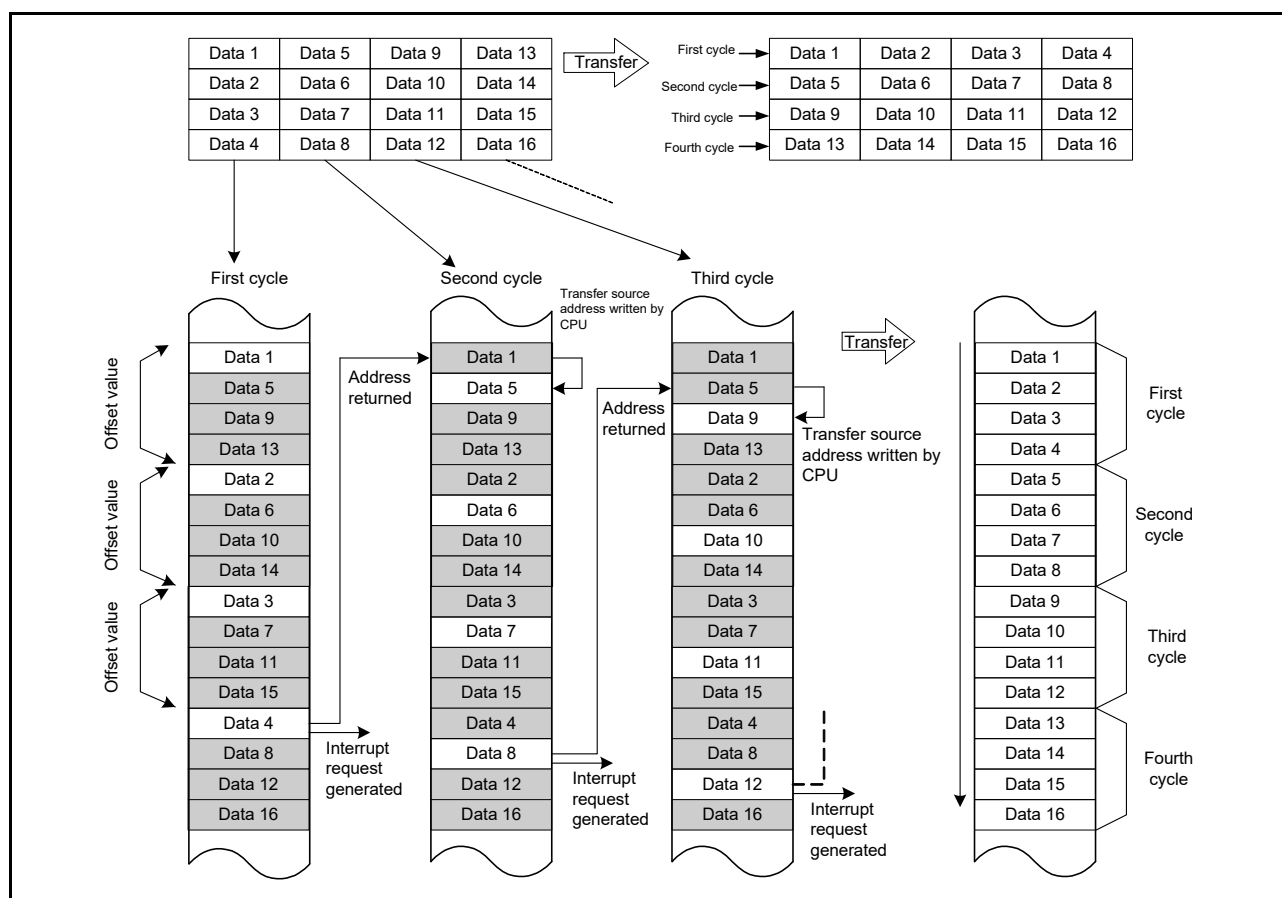


Figure 18.8 XY Conversion Operation Using Offset Addition in Repeat Transfer Mode

When a transfer starts, the offset value is added to the transfer source address every time data is transferred. The transfer data is written to the destination continuous addresses. When data 4 is transferred, which means that the repeat size of transfers is completed, the transfer source address returns to the transfer start address (address of data 1 on the transfer source) and a repeat size end interrupt is requested. While this interrupt stops the transfer temporarily, perform the following.

- DMAC0.DMSAR register: Rewrite the DMA transfer source address to the address of data 5 (with the above example, the data 1 address + 4).
- DMAC0.DMCNT register: Set the DTE bit to 1.

The DMA transfer is resumed from the state when the DMA transfer is stopped. After that, the operations described above are repeated until the transfer source data is transposed to the destination area (XY conversion).

Figure 18.9 shows a flowchart of the XY conversion.

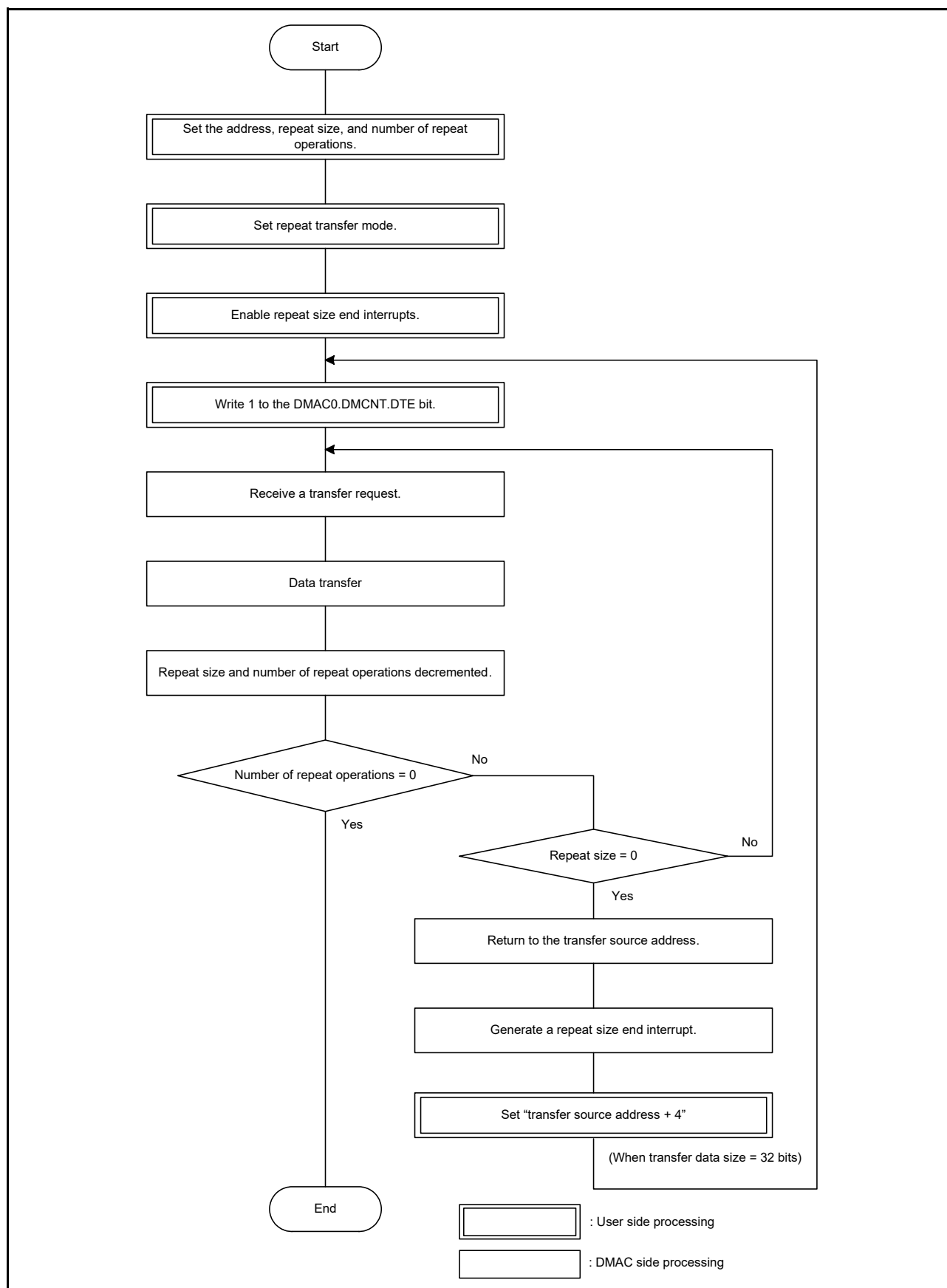


Figure 18.9 XY Conversion Flowchart Using Offset Addition in Repeat Transfer Mode

18.3.4 Request Sources

Software, the interrupt requests from the peripheral modules, and the external interrupt requests can be specified as the DMA request sources. Setting the DMACm.DMTMD.DCTG[1:0] bits selects the request source.

(1) Trigger by Software

Setting the DMACm.DMTMD.DCTG[1:0] bits to 00b enables the trigger by software.

To start DMA transfer by software, set the DMACm.DMTMD.DCTG[1:0] bits to 00b, and then set the DMACm.DMCNT.DTE bit to 1 (DMA transfer is enabled) and the DMACm.DMREQ.SWREQ bit to 1 (DMA transfer is requested) with the DMAST.DMST bit set to 1 (DMAC module start).

When the DMAC is triggered by software while the DMACm.DMREQ.CLRS bit is 0, the DMACm.DMREQ.SWREQ bit is set to 0 after data transfer is started in response to a DMA transfer request.

When the DMAC is triggered by software while the CLRS bit is 1, the SWREQ bit is not set to 0 after data transfer is started. In this case, a DMA transfer request is issued again after completion of a transfer.

(2) Trigger by Interrupt Requests from On-Chip Peripheral Modules or External Interrupt Requests

Interrupt requests from the on-chip peripheral modules and external interrupt requests can be specified as the DMA request sources. The request source can be selected separately for each channel using the ICU.DMRSRm registers (m = 0 to 7).

The DMA transfer is triggered when an interrupt request from the on-chip peripheral module or an external interrupt request is generated while the DMACm.DMTMD.DCTG[1:0] bits are set to 01b (interrupts from the peripheral modules and the external interrupt pins are selected), the DMACm.DMCNT.DTE bit is set to 1 (DMA transfer is enabled), and the DMAST.DMST bit is set to 1 (DMAC module start).

For interrupt requests specified as DMA request sources, refer to Table 15.5, Interrupt Vector Table, in section 15, Interrupt Controller (ICUD).

18.3.5 Operation Timing

Figure 18.10 and Figure 18.11 show DMAC operation timing examples.

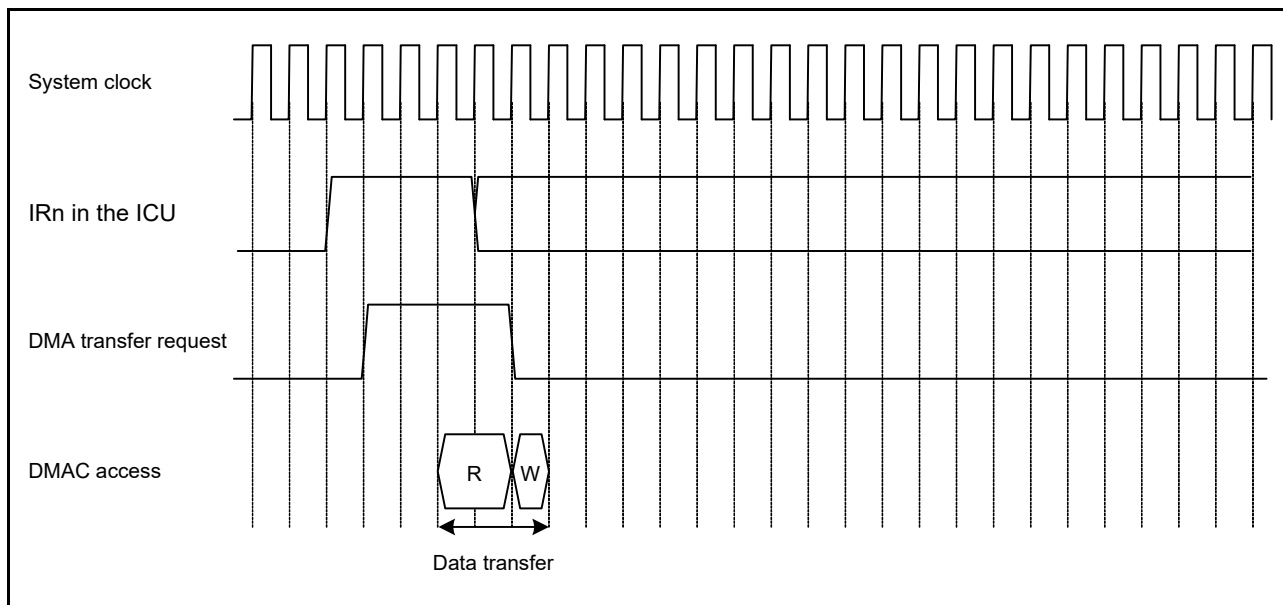


Figure 18.10 DMAC Operation Timing Example (1) (Trigger by Interrupt from Peripheral Module/External Interrupt Input Pin, Normal Transfer Mode, Repeat Transfer Mode)

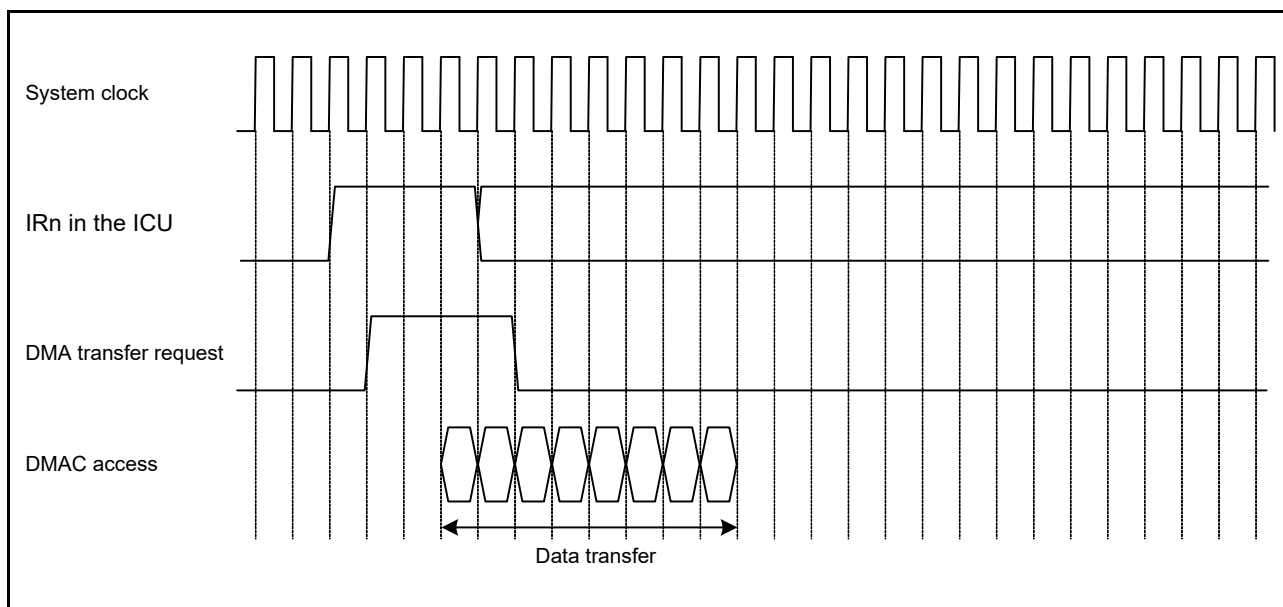


Figure 18.11 DMAC Operation Timing Example (2) (Trigger by Interrupt from Peripheral Module/External Interrupt Input Pin, Block Transfer Mode, Block Size = 4)

18.3.6 DMAC Execution Cycles

Table 18.7 lists execution cycles in one DMAC data transfer operation.

Table 18.7 DMAC Execution Cycles

Transfer Mode	Data Transfer (Read)	Data Transfer (Write)
Normal	Cr+1	Cw
Repeat	Cr+1	Cw
Block*1	P × Cr	P × Cw

Note 1. This is the case when the block size is 2 or more. When the block size is 1, normal transfer cycle is applied.

P: Block size (DMCRAH register setting)

Cr: Data read destination access cycle

Cw: Data write destination access cycle

Cr and Cw depend on the access destination. For the number of cycles for each access destination, refer to section 60, RAM, section 62, Flash Memory (FLASH), section 5, I/O Registers, and section 16.2.6, External Bus.

The unit for +1 in “Data Transfer (Read)” column is one system clock cycle (ICLK).

For the operation example, refer to section 18.3.5, Operation Timing.

18.3.7 Activating the DMAC

Figure 18.12 shows the register setting procedure.

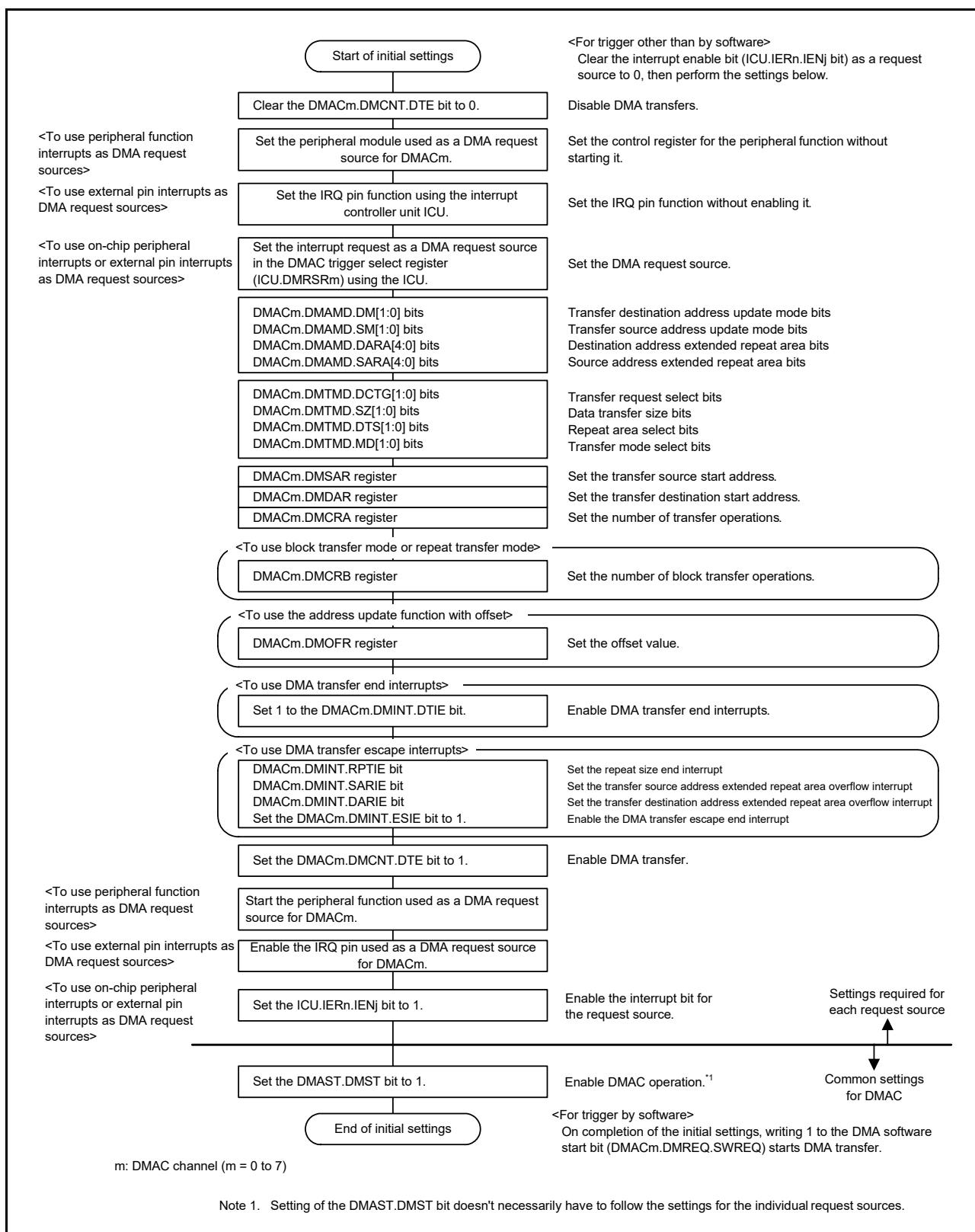


Figure 18.12 Register Setting Procedure

18.3.8 Starting DMA Transfer

Setting the DMACm.DMCNT.DTE bit to 1 (DMA transfer enabled) and setting the DMAST.DMST bit to 1 (DMAC module start) enable DMA transfer of channel m (m = 0 to 7).

Another transfer request cannot be accepted during the transfer of other DMAC channel or DTC. When the proceeding transfer is completed, channel arbitration is performed where a DMA transfer request of the highest priority channel is accepted and DMA transfer of the channel starts. When DMA transfer starts, the DMACm.DMSTS.ACT flag is set to 1 (the DMAC is in the active state).

18.3.9 Registers during DMA Transfer

The DMAC registers are updated by a DMA transfer. The value to be updated differs according to the other settings and the transfer state. The registers to be updated are DMSAR, DMDAR, DMCRA, DMCRB, DMCNT, and DMSTS of DMACm.

(1) DMA Source Address Register (DMACm.DMSAR)

When data has been transferred in response to one transfer request, the contents of this register are updated to the address to be accessed by the next transfer request.

For details on register update operation in each transfer mode, refer to Table 18.3 to Table 18.5.

(2) DMA Destination Address Register (DMACm.DMDAR)

When data has been transferred in response to one transfer request, the contents of this register are updated to the address to be accessed by the next transfer request.

For details on register update operation in each transfer mode, refer to Table 18.3 to Table 18.5.

(3) DMA Transfer Count Register (DMACm.DMCRA)

When data has been transferred in response to one transfer request, the count value is updated. The update operation depends on the transfer mode selected.

For details on register update operation in each transfer mode, refer to Table 18.3 to Table 18.5.

(4) DMA Block Transfer Count Register (DMACm.DMCRB)

When data has been transferred in response to one transfer request, the count value is updated. The update operation depends on the transfer mode selected.

For details on register update operation in each transfer mode, refer to Table 18.3 to Table 18.5.

(5) DMA Transfer Enable Bit (DMACm.DMCNT.DTE)

Although the DMACm.DMCNT.DTE bit enables or disables data transfer by the register write access, it is automatically set to 0 by the DMAC according to the DMA transfer state.

The conditions for clearing this bit by the DMAC are as follows:

- When the specified total volume of data transfer is completed
- When DMA transfer is stopped by the repeat size end interrupt
- When DMA transfer is stopped by the extended repeat area overflow interrupt

Writing to the registers for the channels when the corresponding DMACm.DMCNT.DTE bit is set to 1 is prohibited (except for the DMACm.DMCNT register). In this case, writing must be performed after the bit is set to 0.

(6) DMA Active Flag (DMACm.DMSTS.ACT)

The DMACm.DMSTS.ACT flag indicates whether the DMACm is in the idle or active state.

This flag is set to 1 when the DMAC starts data transfer, and is set to 0 when data transfer in response to one transfer request is completed.

Even when DMA transfer is stopped by writing 0 to the DMACm.DMCNT.DTE bit during DMA transfer, this flag remains 1 until DMA transfer is completed.

(7) Transfer End Interrupt Flag (DMACm.DMSTS.DTIF)

The DMACm.DMSTS.DTIF flag is set to 1 after DMA transfer of the total transfer size of data is completed.

When both this flag and the DMACm.DMINT.DTIE bit are set to 1, a transfer end interrupt is requested.

This flag is set to 1 when the DMA transfer bus cycle is completed and the DMACm.DMSTS.ACT flag is set to 0 indicating the DMA transfer end.

This flag is automatically set to 0 when the DMACm.DMCNT.DTE bit is set to 1 during the interrupt handling.

(8) Transfer Escape End Interrupt Flag (DMACm.DMSTS.ESIF)

The DMACm.DMSTS.ESIF flag is set to 1 when a repeat size end interrupt or extended repeat area overflow interrupt is requested. When this bit and the DMACm.DMINT.ESIE bit are set to 1, a transfer escape end interrupt is requested.

This flag is set to 1 when the bus cycle of the DMA transfer having caused the interrupt request is completed and the DMACm.DMSTS.ACT flag is set to 0 indicating the DMA transfer end.

This flag is automatically set to 0 when the DMACm.DMCNT.DTE bit is set to 1 during an interrupt handling.

Before sending an interrupt request from the DMAC to the CPU or the DTC, the interrupt control register must be set.

For details, refer to section 15, Interrupt Controller (ICUD).

18.3.10 Channel Priority

When multiple DMA transfer requests are present, the DMAC determines the priority of channels that have DMA transfer requests.

The channel priority is fixed as channel 0 > channel 1 > channel 2 > channel 3 > channel 4 > channel 5 > channel 6 > channel 7 (channel 0: highest).

When a DMA transfer request is generated during data transfer, channel arbitration is started after the final data has been transferred, and DMA transfer of the higher-priority channel starts.

18.4 Ending DMA Transfer

The operation for ending DMA transfer depends on the transfer end conditions. When DMA transfer ends, the DMCNT.DTE bit and the DMSTS.ACT flag of DMACm are changed from 1 to 0, indicating that DMA transfer has ended.

18.4.1 Transfer End by Completion of Specified Total Number of Transfer Operations

(1) In Normal Transfer Mode (DMACm.DMTMD.MD[1:0] = 00b)

When the value of the DMACm.DMCRA register changes from 1 to 0, DMA transfer ends on the corresponding channel, and the DMACm.DMCNT.DTE bit is set to 0 and the DMACm.DMSTS.DTIF flag is set to 1 at the same time. If the DMACm.DMINT.DTIE bit is 1 at this time, a transfer end interrupt request is issued to the CPU or the DTC.

(2) In Repeat Transfer Mode (DMACm.DMTMD.MD[1:0] = 01b)

When the value of the DMACm.DMCRA register changes from 1 to 0, DMA transfer ends on the corresponding channel, and the DMACm.DMCNT.DTE bit is set to 0 and the DMACm.DMSTS.DTIF flag is set to 1 at the same time. If the DMACm.DMINT.DTIE bit is 1 at this time, an interrupt request is issued to the CPU or the DTC.

(3) In Block Transfer Mode (DMACm.DMTMD.MD[1:0] = 10b)

When the value of the DMACm.DMCRA register changes from 1 to 0, DMA transfer ends on the corresponding channel, and the DMACm.DMCNT.DTE bit is set to 0 and the DMACm.DMSTS.DTIF bit is set to 1 at the same time. If the DMACm.DMINT.DTIE bit is 1 at this time, an interrupt request is issued to the CPU or the DTC.

Before sending an interrupt request from the DMAC to the CPU or the DTC, the interrupt control register must be set. For details, refer to [section 15, Interrupt Controller \(ICUD\)](#).

18.4.2 Transfer End by Repeat Size End Interrupt

In repeat transfer mode, a repeat size end interrupt is requested when transfer of a 1-repeat size of data is completed while the DMACm.DMINT.RPTIE bit is set to 1. When the interrupt is requested to complete DMA transfer, the DMACm.DMCNT.DTE bit is set to 0 and the DMACm.DMSTS.ESIF flag is set to 1. If the DMACm.DMINT.ESIE bit is 1 at this time, an interrupt request is issued to the CPU or the DTC. Here, the transfer can be resumed by writing 1 to the DMACm.DMCNT.DTE bit.

A repeat size end interrupt can be requested also in block transfer mode. In block transfer mode, the interrupt is requested in the same way as in repeat transfer mode when transfer of a 1-block size data is completed.

Before sending an interrupt request from the DMAC to the CPU or the DTC, the interrupt control register must be set. For details, refer to [section 15, Interrupt Controller \(ICUD\)](#).

18.4.3 Transfer End by Interrupt on Extended Repeat Area Overflow

When an overflow on the extended repeat area occurs while the extended repeat area is specified and the DMINT.SARIE or DMINT.DARIE bit of DMACm is set to 1, an interrupt by an extended repeat area overflow is requested. When the interrupt is requested, the DMA transfer is terminated, the DMACm.DMCNT.DTE bit is set to 0, and the DMACm.DMSTS.ESIF flag is set to 1. If the DMACm.DMINT.ESIE bit is 1 at this time, an interrupt request is issued to the CPU or the DTC.

Even if an interrupt by an extended repeat area overflow is requested during a read cycle, the following write cycle is performed.

In block transfer mode, even if an interrupt by an extended repeat area overflow is requested during a 1-block transfer, the remaining data in the block is transferred; transfer is terminated after a block transfer.

Before sending an interrupt request from the DMAC to the CPU or the DTC, the interrupt control register must be set. For details, refer to [section 15, Interrupt Controller \(ICUD\)](#).

18.5 Interrupts

Each DMAC channel can output an interrupt request to the CPU or the DTC after transfer in response to one request is completed. When the transfer destination is the external bus or the on-chip peripheral bus, an interrupt request is generated upon completion of data write to the write buffer not to the actual transfer destination.

Table 18.8 lists the relation among the interrupt sources, the interrupt status flags, and the interrupt enable bits. Figure 18.13 shows the schematic logic diagram of interrupt outputs (DMAC0 to DMAC3). Figure 18.14 shows the schematic logic diagram of interrupt outputs (DMAC4 to DMAC7). Figure 18.15 shows the DMAC interrupt handling routine to resume/terminate DMA transfer.

Table 18.8 Relation among Interrupt Sources, Interrupt Status Flags, and Interrupt Enable Bits

Interrupt Sources		Interrupt Enable Bits	Interrupt Status Flags	Request Output Enable Bits
Transfer end		—	DMACm.DMSTS.DTIF	DMACm.DMINT.DTIE
Escape transfer end	Repeat size end	DMACm.DMINT.RPTIE	DMACm.DMSTS.ESIF	DMACm.DMINT.ESIE
	Source address extended repeat area overflow	DMACm.DMINT.SARIE		
	Destination address extended repeat area overflow	DMACm.DMINT.DARIE		

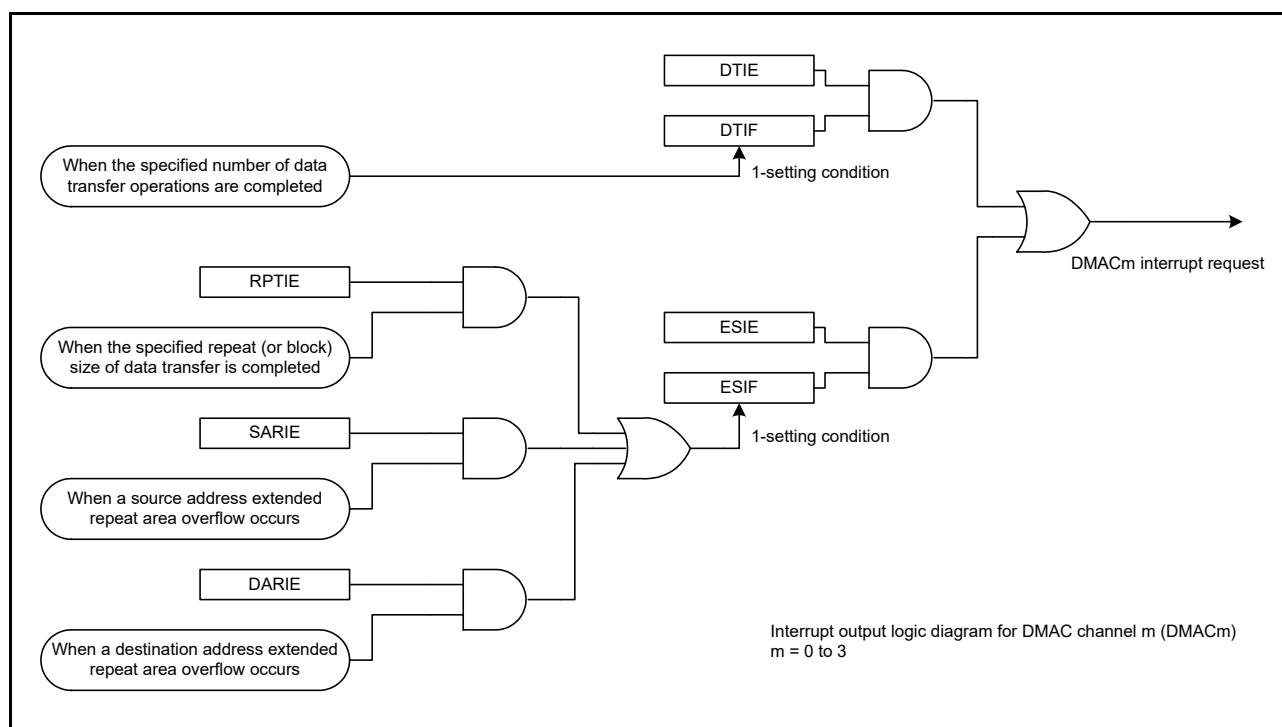


Figure 18.13 Schematic Logic Diagram of Interrupt Outputs (DMAC0 to DMAC3)

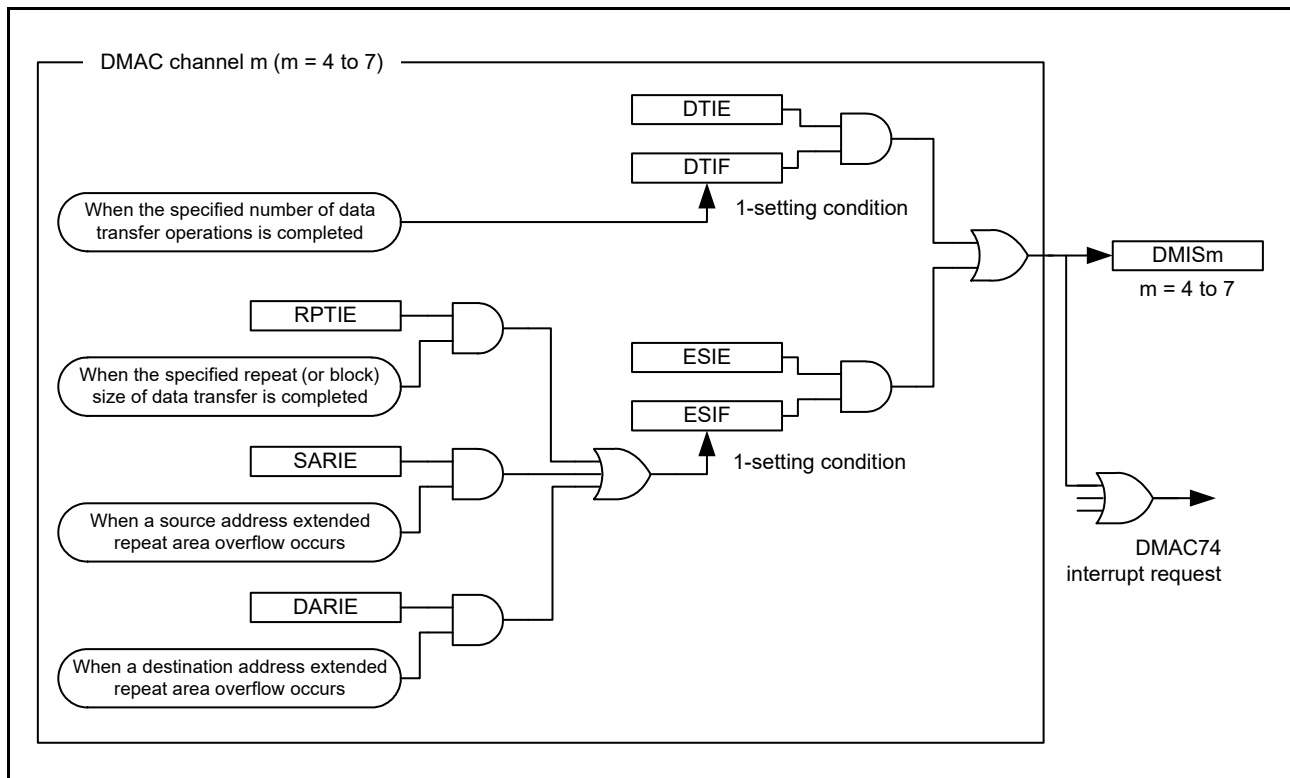


Figure 18.14 Schematic Logic Diagram of Interrupt Outputs (DMAC4 to DMAC7)

Specifically, the different procedures are used for canceling an interrupt to restart DMA transfer in the following two cases: (1) discontinuing or terminating DMA transfer and (2) continuing DMA transfer.

(1) When Discontinuing or Terminating DMA Transfer

Write 0 to the DMACm.DMSTS.DTIF flag to clear a transfer end interrupt, and to the DMACm.DMSTS.ESIF flag to clear a repeat size interrupt and an extended repeat area overflow interrupt. The DMACm remains in the stop state. When starting another DMA transfer after that, set the appropriate registers, and set the DMACm.DMCNT.DTE bit to 1 (DMA transfer enabled).

(2) When Continuing DMA Transfer

Write 1 to the DMACm.DMCNT.DTE bit. The DMACm.DMSTS.ESIF flag is automatically set to 0 (interrupt source cleared), and DMA transfer is resumed.

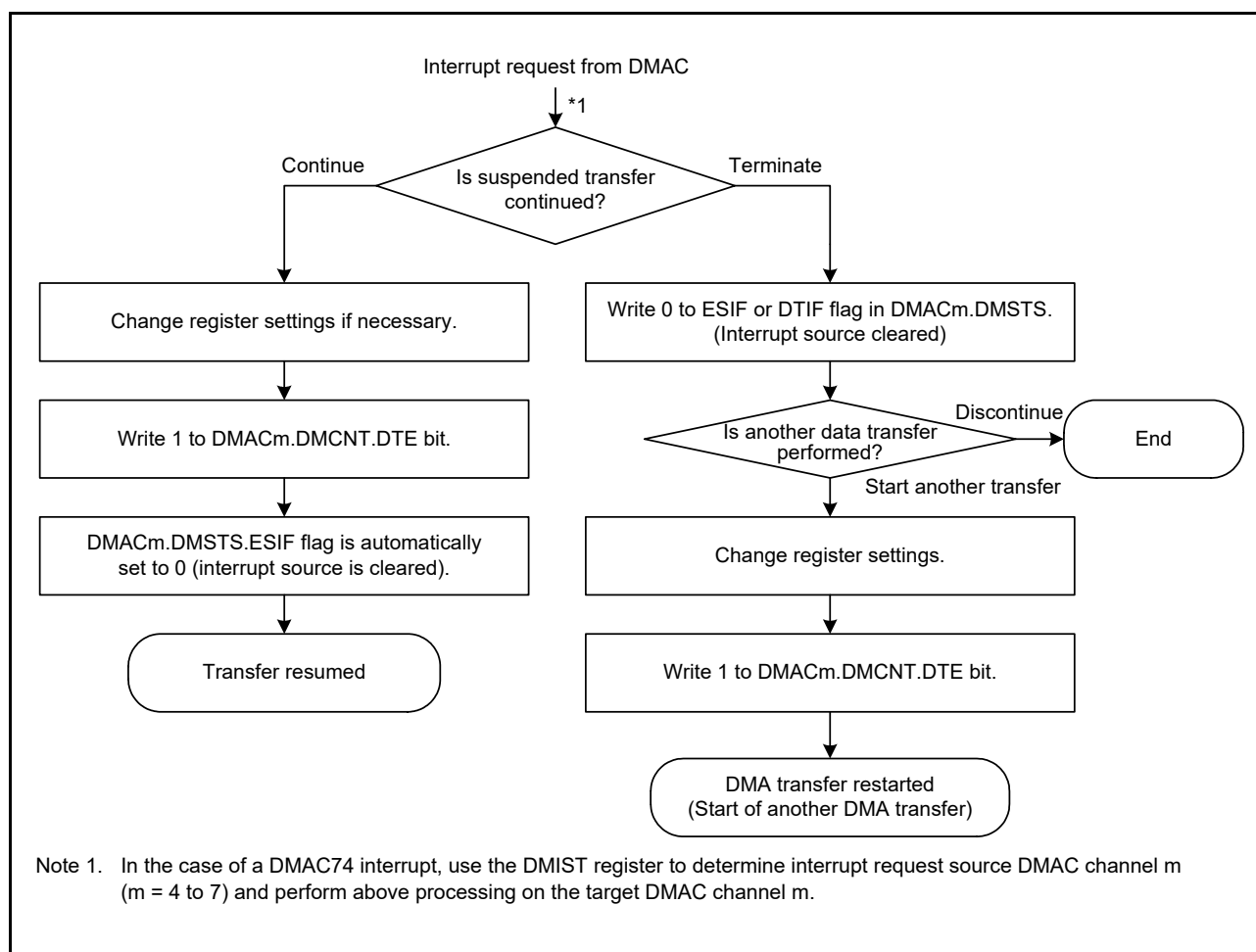


Figure 18.15 DMAC Interrupt Handling Routine to Resume/Terminate DMA Transfer

18.6 Event Link

Each DMAC channel outputs an event link request signal each time the channel completes data transfer (or block transfer in block transfer mode). However, when the transfer destination is the external bus or internal peripheral bus, an event link request signal is generated when the writing to the write buffer is accepted.

18.7 Low-Power Consumption Function

Before transition to the module-stop state, all-module clock stop mode, software standby mode, or deep software standby mode, set the DMAST.DMST bit to 0 (the DMAC suspended), and then perform the following.

(1) Module-Stop Function

Writing 1 to the MSTPCRA.MSTPA28 bit (transition to the module-stop state) enables the module-stop function of the DMAC. If DMA transfer is in progress at the time a 1 is written to the MSTPA28 bit, the transition to the module-stop state proceeds after DMA transfer has ended. While the MSTPA28 bit is 1, accessing the DMAC registers are prohibited. Writing 0 to the MSTPA28 bit releases the DMAC from the module-stop state.

(2) All-Module Clock Stop Mode

Make settings in accord with the procedure under section 11.6.2.1, Transition to All-Module Clock Stop Mode, in section 11, Low Power Consumption.

If DMA transfer operations are in progress at the time the WAIT instruction is executed, the transition to all-module clock stop mode follows the completion of DMA transfer.

The DMAC is released from the module-stop state by writing 0 to the MSTPCRA.MSTPA28 bit following recovery from all-module clock stop mode.

(3) Software Standby and Deep Software Standby Modes

Make settings in accord with the procedure under section 11.6.3.1, Transition to Software Standby Mode or section 11.6.4.1, Transition to Deep Software Standby Mode, in section 11, Low Power Consumption.

If DMA transfer operations are in progress at the time the WAIT instruction is executed, the transition to software standby or deep software standby follows the completion of DMA transfer.

(4) Note on Low-Power Consumption Function

For the WAIT instruction and the register setting procedure, refer to section 11.7.6, Timing of WAIT Instructions in section 11, Low Power Consumption.

To perform DMA transfer after returning from low-power consumption mode, set the DMAST.DMST bit to 1 again.

To use a request that is generated in all-module clock-stop mode and software standby mode as an interrupt request to the CPU but not as a DMAC startup request, specify the CPU as the interrupt request destination in accordance with the description in section 15, Interrupt Controller (ICUD), and then execute the WAIT instruction.

18.8 Usage Notes

18.8.1 DMA Transfer to External Devices

In DMA transfer to an external device, the DMACm.DMSTS.ACT flag may be set to 0 (DMAC transfer suspended) during the period from the beginning of the final data write to the end of the external bus access.

18.8.2 DMA Transfer to Peripheral Modules

In DMA transfer to a peripheral module, the DMACm.DMSTS.ACT flag may be set to 0 (DMAC transfer suspended) during the period from the beginning of the final data write to the end of the peripheral bus access.

18.8.3 Access to the Registers during DMA Transfer

Do not write to the DMSAR, DMDAR, DMCRA, DMCRB, DMTMD, DMINT, DMAMD, DMOFR, and DMCSL registers of DMACm while the DMSTS.ACT flag of the same channel is set to 1 (DMAC active state) or the DMCNT.DTE bit of the same channel is set to 1 (DMA transfer enabled).

18.8.4 DMA Transfer to Reserved Areas

DMA transfer to the reserved areas is prohibited. If such an access is made, transfer results are not guaranteed. For details on the reserved areas, refer to section 4, Address Space.

18.8.5 Interrupt Request by the DMA Request Source Flag Control Register (DMCSL) at the End of Each Transfer

While the DMACm.DMCSL.DISEL bit is 1, an interrupt request is issued to the CPU at the end of each transfer that has been triggered by one DMA request. Unlike the transfer end interrupt that the DMAC outputs or the escape end interrupt, the interrupt of this type is issued to the CPU at the end of DMA transfer without clearing the interrupt status flag of the DMA request source to 0 by changing the interrupt request destination to the CPU. In this case, since the interrupt status flag is not set to 0 at the end of DMAC transfer, it should be set to 0 by the CPU interrupt routine.

The interrupt flag is cleared when the CPU interrupt is accepted.

For the change of the settings on the interrupt flag or the interrupt request destination, refer to section 15, Interrupt Controller (ICUD). For the DMACm.DMCSL.DISEL bit setting, refer to section 18.2.12, DMA Request Source Flag Control Register (DMCSL).

18.8.6 Setting of DMAC Trigger Select Register of the Interrupt Controller (ICU.DMRSRm)

The DMAC trigger select register (ICU.DMRSRm) should be set while the DMA transfer enable bit (DMACm.DMCNT.DTE) is 0 (DMA transfer is disabled). Moreover, the DTC transfer request enable register (ICU.DTCERn) that corresponds to the same vector number that has been set by the ICU.DMRSRm register should not be set to 1. For details on the ICU.DTCERn and ICU.DMRSRm registers, refer to section 15, Interrupt Controller (ICUD).

18.8.7 Suspending or Restarting DMA Transfer

To suspend a DMA transfer request, write 0 to the interrupt enable bit for the request source (ICU.IERn.IENj bit). To restart the DMA transfer, write 1 to the ICU.IERn.IENj bit with the setting shown in section 18.3.7, Activating the DMAC.